

Low-rank neural networks preserve the MLPs Neural Tangent Kernel

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Abstract

We develop a Neural Tangent Kernel (NTK) theory for low-rank random feature networks (RF-LR) that provides a principled, computationally efficient route to the kernel regime. Assuming fixed weights as classical NTK theory predicts, we prove that RF-LR preserves the same reproducing kernel Hilbert space (RKHS) as the shallow ReLU kernel; low-rank weight matrices do not lose expressivity. Combining Fisher–Kibble decoupling of Gaussian process outputs with a hypergeometric analysis of the mean NTK near $\rho = 1$, we establish the Puiseux expansion and RKHS equivalence. We link the $I(r) \sim 1/\sqrt{r}$ decay to tighter concentration as r grows. The empirical three-layer NTK concentrates around its deterministic limit with sub-Gaussian tails in r . At the spectral level, the NTK Gram matrix exhibits a spiked and shifted Marchenko–Pastur law. We give an explicit NTK recursion and closed-form depth expansion. Taken together, these results establish that RF-LR preserves expressivity and gives optimization guarantees while lowering the parameter cost from $O(N^2)$ to $O(rN)$.

Keywords: Neural Tangent Kernel, Random Features, Low Rank, Scaling Laws, Infinite Width, Marchenko–Pastur

1. Introduction

Multi-Component and Multi-Layer Neural Networks (MMNNs), also called low-rank random-feature networks (RF-LR), are architectures where each layer is implemented as a random feature map followed by a trainable linear combination with a low-dimensional bottleneck [Zhang et al. \(2025\)](#). Concretely, for each layer we sample frozen random feature directions $w^{(\ell)}$ and train only the readout weights $A^{(\ell)}$, which map the feature vector into a bottleneck of dimension r . This reduces the trainable parameter count from $O(N^2)$ (dense layers) to $O(rN)$ while keeping the random-feature geometry explicit. Recent empirical work by [Zhang et al. \(2025\)](#) has demonstrated that RF-LRs achieve global strong-convergence across a wide range of tasks, including neural-network based PDE solving with and without Physics-Informed losses [Raissi et al. \(2017\)](#). Empirically, training can display stepwise dynamics (plateaus separated by sharp drops) in some regimes [Zhang et al. \(2025\)](#). This paper does not aim to explain these phenomena beyond the kernel regime; instead we develop an NTK-level theory for RF-LR at initialization and in the lazy/NTK regime, focusing on the induced RKHS, concentration in the bottleneck dimension r , and spectral predictions for the Gram matrix.

The Neural Tangent Kernel (NTK) framework [Jacot et al. \(2018\)](#) provides the natural lens for analyzing the early training stage. This perspective has been extended to finite width [Huang and Yau \(2019\)](#); [Dyer and Gur-Ari \(2019\)](#) and large-depth [Guillen et al. \(2025\)](#); [Yang et al. \(2023\)](#); [Terjék \(2022\)](#), revealing how the Laplacian kernel structure governs the gradient descent (GD) training dynamics and generalization. Applying NTK theory to RF-LR raises specific technical questions: the kernel recursion differs from the standard MLP recursion because only $A^{(\ell)}$ is trained, and the bottleneck dimension r induces additional concentration and scaling effects.

We show that, in the three-layer RF-LR setting under homogeneity and isotropy assumptions, the *mean* NTK induces the same RKHS as the shallow ReLU kernel (hence depth does not enlarge the RKHS in this regime), while requiring only $O(rN)$ trainable parameters.

To keep the analysis explicit, we work under standard simplifying assumptions (ReLU homogeneity, isotropic Gaussian features, and a sequential infinite-width limit). In the NTK regime, the kernel is (approximately) constant along training [Jacot et al. \(2018\)](#); accordingly, we analyze the induced kernel at initialization and its deterministic limits.

Scope and guarantees (what we analyze and what we do not claim).

- **Objects analyzed.** We study (i) the RF-LR NTK recursion in the sequential infinite-width limit (Theorem 3); (ii) the deterministic/mean kernels that arise at initialization (including the three-layer mean kernel used for the RKHS analysis); and (iii) the empirical Gram matrix at initialization in a random-matrix scaling regime, including its spike–bulk spectral limit (Section ??).
- **Optimization in-scope.** In the lazy/NTK regime, gradient flow reduces to kernel regression with a fixed kernel (Theorem 3). Convergence rates are governed by the spectrum of the (centered) Gram matrix on the training set; our contribution is to provide explicit kernel formulas/limits and spectral predictions in the RF-LR setting.
- **Not claimed.** We do not analyze feature learning (time evolution of the NTK), adaptive optimizers, or continual-learning/forgetting phenomena. Any references to such effects are purely motivational and are not used as claims or evidence.

Low rank architecture

The considered architecture is a stacking of high then low width layers:

The loss landscape of neural networks is closely related to generalization and trainability, and in the kernel regime the spectral properties of the Gram matrix control optimization rates and spectral bias [Jacot et al. \(2018\)](#); [Cao et al. \(2020\)](#). We focus on this kernel-regime viewpoint for RF-LR by characterizing the limiting kernels and their spectra at initialization under explicit assumptions.

Our approach develops an NTK theory specialized to intra-layers random variables output arising after large-layer acting as gaussian processes,

This work addresses the following fundamental questions :

- *How Low-Rank preserves MLPs expressivity and how random features simplify NTK analysis ?*

This question is closely related to the open question of the curse of dimensionality in deep learning optimization [Na and Yang \(2025\)](#). In particular we show that the curse of dimensionality occurring in the NTK exponential-in-dim spectral decay is not mitigated by low-rank structure.

By exploiting particularities from Fisher and Kibble distributions [Fisher \(1915\)](#); [Hotelling \(1953\)](#) and manipulation of non-integer Puiseux polynomials series of correlation-kernels inside expectations, we are able to derive the RKHS structure inherent in frozen weights using a characterization from [Bietti and Bach \(2021\)](#). We focus specifically on the NTK regime where analytical tractability is greatest and the advantages of the low-rank random feature structure are most clearly visible.

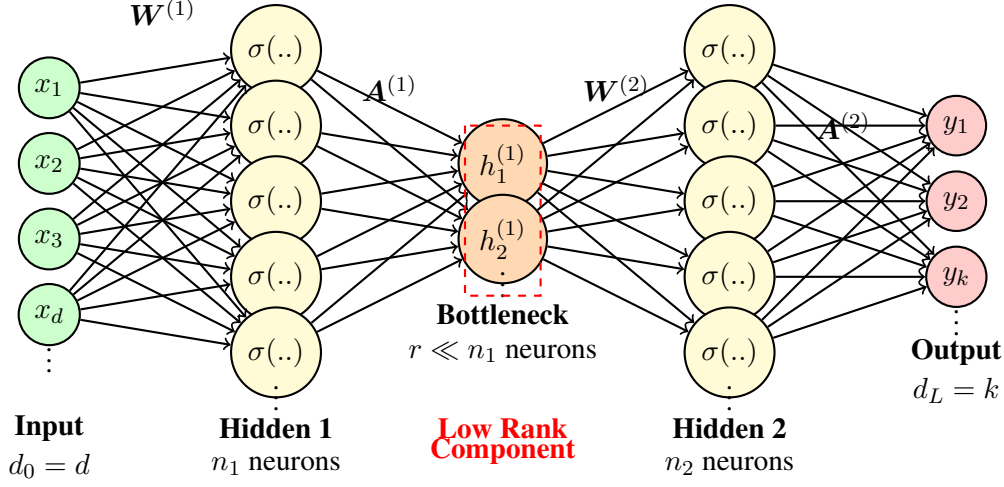


Figure 1: Architecture of a Multi-component Multi-layer Neural Network (RF-LR). The red dashed connections indicate the low-rank component that reduces dimensionality from n_1 to $r \ll n_1$ neurons. We **only train the output weights** $A^{(1)}$ and $A^{(2)}$.

Our setup also clarifies which parts of finite-width NTK corrections simplify when feature weights are frozen: cross-layer dependencies arising from jointly trained weights are absent, so one can isolate the remaining sources of randomness (notably, the rank- r bottleneck). We do not develop a full finite-width correction theory here, but the explicit recursion we derive provides a starting point for such analyses.

By characterizing the NTK recursion, RKHS geometry, and Gram-matrix spectrum under explicit assumptions, we provide kernel-regime predictions for RF-LR that can be compared to empirical behavior. Extensions beyond the lazy regime (e.g., mean-field or feature-learning analyses) are left for future work.

1.1. Related work

The Neural Tangent Kernel (NTK) framework [Jacot et al. \(2018\)](#) characterizes infinitely wide neural networks trained by gradient descent as kernel methods with a fixed kernel. This framework has been extended to finite width [Arora et al. \(2019a\)](#), convolutional architectures [Arora et al. \(2019b\)](#), and depth [Yang \(2019\)](#); [Terjék \(2022\)](#), establishing a comprehensive theoretical foundation for understanding wide network training. Recent work by [Benigni and Paquette \(2025\)](#) characterizes the eigenvalue distribution of the NTK in the quadratic scaling regime (width $N \sim d^2$) for single-layer networks, establishing deformed Marchenko–Pastur laws with spike-bulk structure. Their analysis requires careful tracking of correlation diagrams and moment calculations. We extend this spectral analysis to the three-layer case with random features, where the rank- r bottleneck and frozen feature weights fundamentally alter the kernel’s spectral properties. A key technical advantage of our approach is that we **avoid computing correlation diagrams** by leveraging the decoupling properties of Fisher–Kibble distributions and the independence structure inherent in frozen random features,

which simplifies the analysis while maintaining rigor. Parallel to this, random feature approximations [Rahimi and Recht \(2007, 2008\)](#) provide a practical route to kernel approximation by sampling features from suitable distributions, with rigorous guarantees for generalization [Bach \(2017\)](#); [Rudi and Rosasco \(2017\)](#) and approximation [Sriperumbudur et al. \(2015\)](#). Notably, freezing random features ensures full support for the Barron-space limit in the mean-field regime throughout training, so universal approximation persists at all finite times—a property stronger than classical kernel ridge assumptions. NTK spectrum dependence with depth at the edge of chaos has been analyzed for MLPs in [Terjék and González-Sánchez \(2025\)](#). Our RF-LR setting has a different recursion and, in particular, the bottleneck scaling can lead to kernel saturation and gap decay with depth (Section 4).

Low-rank and factorized architectures [Sainath et al. \(2013\)](#); [Novikov et al. \(2015\)](#); [Arora et al. \(2018\)](#) substantially reduce parameter counts in neural networks. However, the training dynamics of factorized weights can differ from unfactorized gradient descent due to implicit regularization effects. When only output weights are trained while feature weights remain frozen, the architecture avoids Riemannian geometry complications while achieving linear parameter scaling $O(rN)$, representing an extreme case of factorization that enables clean theoretical analysis.

Neural networks exhibit a well-documented spectral bias toward low frequencies [Rahaman et al. \(2019\)](#); [Xu et al. \(2019\)](#), where low-frequency components are learned before high-frequency ones. This bias has been quantified via NTK analysis [Basri et al. \(2020\)](#) and Fourier methods [Tancik et al. \(2020\)](#), revealing how kernel structure shapes learning dynamics. The "deep equals shallow" result for ReLU kernels in the NTK regime [Bietti and Bach \(2021\)](#) establishes that the RKHS induced by deep networks coincides with that of a shallow ReLU kernel, providing formal NTK spectrum decay rates. The same work characterizes the eigenvalue decay of rotation-invariant kernels on the sphere via endpoint expansions of the kernel $\kappa(\rho)$ near $\rho = \pm 1$, relating the smoothness properties of the kernel at the endpoints to the decay rate of spherical-harmonic eigenvalues. This connection between kernel smoothness and spectral decay provides a principled framework for understanding how RKHS structure, kernel spectra, and training behavior interact.

Multi-index approaches [Ben Arous et al. \(2021\)](#); [Damian et al. \(2022\)](#); [Abbe et al. \(2023\)](#) learn functions of the form $f(x) = \sum_{j=1}^r g_j(\langle v_j, x \rangle)$ by discovering latent directions sequentially, with training displaying staircase dynamics featuring plateaus for inactive directions. In extensive-rank regimes $r \asymp d^\beta$, these works derive scaling laws [Damian et al. \(2022\)](#) that connect rank, dimension, and sample complexity. three-layer RF-LR sees their mid layer acting as having r independent functions (approximated by the random features basis) forming a multi-index model in function space. Those functions only interact through reconstructing output functions in next layers. This point of view is a novel way to understand low rank weight matrices provided that you do not train the random features to ensure approximation of dictionary - reconstruction - approxim .. etc ..

From a data-dependent scaling perspective, achieving and maintaining NTK behavior in fully-trainable deep networks can impose stringent width–data requirements, often quoted in the literature as polynomial $O(N^p)$ in the width N , with worst-case exponents $p = 4$, and best-case exponents $p = 1$ [Nguyen et al. \(2022\)](#) for quadratic scaling. To ensure the NTK has a positive smallest eigenvalue—crucial for optimization convergence—at least one layer must scale with the number of data samples n , while other layers can scale arbitrarily up to logarithmic factors for deep networks, motivating architectures that achieve NTK regime behavior with tractable computational requirements for large scale applications.

Our Contributions

Our contributions are: (i) an NTK recursion specialized to RF-LR together with rank-driven concentration bounds, (ii) a three-layer spectral theory yielding a signal–noise decomposition of the empirical kernel and a deformed Marchenko–Pastur law with a spike for the uncentered Gram matrix spectrum, and (iii) a rigorous proof—grounded in [Bietti and Bach \(2021\)](#)—that the three-layer mean NTK induces the same RKHS as the shallow ReLU kernel (extension to general depth is stated as an open problem/conjecture).

More specifically:

- **NTK recursion** We derive the new NTK recursion expansion for RF-LR ($\sigma_A, \sigma_c, \beta = 1$), yielding clean expressions for three-layer and general L -layer networks.
- **RKHS Preserved (three-layer case)** In the NTK regime $f_t(x) \approx f_0(x) + \langle \nabla_\theta f_0(x), \theta_t - \theta_0 \rangle$ and $f_t \in \mathcal{H}_\Theta$ with $\|f\|_{\mathcal{H}_\Theta}^2 = \langle f, \Theta^{-1} f \rangle_{L^2}$. For the three-layer RF-LR, we prove that the mean NTK induces the same RKHS as the shallow ReLU kernel (same spectral decay rate); RF-LR preserves universal approximation (frozen Gaussian $w^{(\ell)}$), the rank- r bottleneck acts as a spectral filter, and the NTK-RKHS uses $O(rN)$ rather than $O(N^2)$ parameters. The extension to general depth $L \geq 3$ is conjectured.
- **Marchenko-Pastur like Gram spectrum (three-layer case).** We state and analyze Fisher–Kibble decoupling [Fisher \(1915\)](#); [Hotelling \(1953\)](#), a signal–noise decomposition of the empirical NTK, and a deformed Marchenko–Pastur law (with spike) for the kernel matrix spectrum under standard random matrix scaling.

2. Network definition and EOC parameterization

2.1. RF-LR architecture

Let $\mathbf{h}^{(0)}(x) = x \in \mathbb{R}^{d_0}$. For $\ell = 1, \dots, L$,

$$\mathbf{h}^{(\ell)}(x) = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \sigma\left(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x) + b_j^{(\ell)}\right) + c. \quad (1)$$

Training policy: $A^{(\ell)}$ and the scalar output bias c are trained; $w^{(\ell)}$ and activation biases $b^{(\ell)}$ are frozen random draws (i.i.d. Gaussian). The $1/\sqrt{n_\ell}$ scaling (NTK parameterization) ensures a well-defined infinite-width limit.

Width vs bottleneck dimension. We distinguish the feature width n_ℓ (number of random features) from the output dimension $d_\ell = \dim(\mathbf{h}^{(\ell)})$. The trainable matrix $A^{(\ell)} \in \mathbb{R}^{d_\ell \times n_\ell}$ has columns $A_j^{(\ell)} \in \mathbb{R}^{d_\ell}$. A *bottleneck* layer corresponds to $d_\ell = r \ll n_\ell$, yielding $O(rn_\ell)$ trainable parameters and $\text{rank}(A^{(\ell)}) \leq r$ automatically.

Training parameters, biases, and initialization. Initialization is

$$w_j^{(\ell)} \sim \mathcal{N}(0, I_{d_{\ell-1}}/d_{\ell-1}), \quad b_j^{(\ell)} \sim \mathcal{N}(0, \beta^2), \quad A_{ij}^{(\ell)} \sim \mathcal{N}(0, \sigma_A^2), \quad c \sim \mathcal{N}(0, \sigma_c^2),$$

We use the *edge-of-chaos (EOC)* scaling for the weights so that $\text{Cov}(w_j^{(\ell)}) = I_{\mathbf{d}_{\ell-1}}/\mathbf{d}_{\ell-1}$ Hayou et al. (2018). In our bottleneck regime we take $\mathbf{d}_{\ell-1} = r$ for $\ell \geq 2$, so $w_j^{(\ell)} \in \mathbb{R}^r$ with covariance I_r/r .

We include an activation bias $b^{(\ell)}$ and a scalar output bias c for completeness, but we avoid introducing extra scaling parameters for them. Output scaling can be absorbed into the learning rate; the mean-field and NTK structures remain qualitatively identical, and the Fourier-space interpretation and stepwise loss phenomenon depend on kernel geometry, not absolute scales.

With this setup in place, we now recall a minimal assumption on the activation that we will use throughout.

Activation homogeneity. We only assume a positively 1-homogeneous nonlinearity σ for Fisher–Kibble decoupling (see Lemma 8):

$$\sigma(\alpha u) = \alpha \sigma(u) \quad \text{for } \alpha \geq 0, u \in \mathbb{R}.$$

This yields the scaling relation $\sigma(\alpha w^\top h) = \alpha \sigma(w^\top h)$ and, away from zero, $\dot{\sigma}(\alpha u) = \dot{\sigma}(u)$.

2.2. Parameterization, signal propagation, and kernels

With the model and initialization fixed, we can now examine parameterization choices, signal propagation at initialization, and the kernel quantities driving our analysis. We begin by tracking the layer-wise variance at initialization.

For general weight variance $\Sigma_w = I_{\mathbf{d}_{\ell-1}}/\mathbf{d}_{\ell-1}$, define the per-layer variance $q^{(\ell)}$. By weight independence we obtain the recursion:

$$q^{(\ell)} = \sigma_c^2 + \frac{\sigma_A^2}{2} q^{(\ell-1)} \text{Tr}(\Sigma_w),$$

This leads to the (EOC) condition:

$$\frac{\sigma_A^2}{2} \text{Tr}(\Sigma_w) = 1.$$

It is chosen to prevent forward activations and backward sensitivities from exploding or vanishing across depth. For example, under $\Sigma_w = I_{\mathbf{d}_{\ell-1}}/\mathbf{d}_{\ell-1}$, one obtains $\sigma_A = \sqrt{2}$ (and we keep σ_c^2 fixed).

Origin of $1/r$ in the NTK recursion. For bottleneck layers $\ell \geq 2$, the input dimension is $\mathbf{d}_{\ell-1} = r$, so EOC prescribes $\Sigma_w = I_r/r$ with $\text{Tr}(\Sigma_w) = 1$. The derivative kernel

$$\dot{\Sigma}^{(\ell)}(x, x') = \mathbb{E}_w [\dot{\sigma}(w^\top h) \dot{\sigma}(w^\top h') \|w\|^2] = \text{Tr}(\Sigma_w) \left(\frac{1}{2} - \frac{\arccos(\rho_\ell)}{2\pi} \right)$$

remains $O(1)$ under EOC. Moreover, with $\text{Tr}(\Sigma_w) = 1$ this quantity is uniformly bounded as

$$0 \leq \dot{\Sigma}^{(\ell)}(\rho) \leq \frac{1}{2}, \quad \forall \rho \in [-1, 1],$$

which can be used as an explicit choice of the constant c_0 in large-depth bounds based on products of derivative kernels. In our RF-LR setting the NTK involves gradients with respect to the bottleneck readouts $A^{(\ell)}$, and the resulting recursion carries an explicit $1/r$ prefactor in the bottleneck regime (Theorem 3). Intuitively, this factor reflects the bottleneck scaling in the parameterization and the

fact that the relevant Jacobian contributions concentrate in an r -dimensional subspace. We provide the full derivation in Appendix A.1.

$$\Theta^{(\ell)} = 1 + \frac{1}{r} \Theta^{(\ell-1)} \cdot \sigma_A^2 \dot{\Sigma}^{(\ell)} + \frac{1}{r} \sigma_A^2 \Sigma^{(\ell)}, \quad \ell \geq 2.$$

In summary: EOC fixes $\text{Tr}(\Sigma_w) = 1$ so that $\dot{\Sigma}^{(\ell)}, \Sigma^{(\ell)}$ are $O(1)$, while the low-rank/bottleneck scaling appears through the explicit prefactor $1/r$ in the NTK recursion.

With an abuse of notation, we denote by $\Sigma^{(\ell)}$ the base kernel and by $\dot{\Sigma}^{(\ell)}$ the derivative kernel. We emphasize that these are not the NNGP kernel and the derivative kernel, but the base kernel and the derivative kernel of the Gaussian process $\mathbf{h}^{(\ell)}$ conditional on $\mathbf{h}^{(\ell-1)}$. By rotation invariance of Gaussian weights, these kernels depend on inputs only through the cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$. We will therefore write $\Sigma^{(\ell)}(\rho)$ and $\dot{\Sigma}^{(\ell)}(\rho)$. When no confusion arises (e.g., for $\ell = 1$), we will also abuse notation and write $\sigma(\rho)$ for the scalar kernel function; this σ denotes the kernel-as-a-function-of- ρ , not the activation σ . We can now state the base and derivative kernels precisely.

Definition 1 (Base and derivative kernels) *In the sequential infinite-width limit, the layer- ℓ base kernel is*

$$\Sigma^{(\ell)}(x, x') = \mathbb{E}_w \left[\sigma(w^\top \mathbf{h}^{(\ell-1)}(x)) \sigma(w^\top \mathbf{h}^{(\ell-1)}(x')) \right], \quad (2)$$

and the derivative kernel is

$$\dot{\Sigma}^{(\ell)}(x, x') = \mathbb{E}_w \left[\dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x)) \dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x')) \|w\|^2 \right]. \quad (3)$$

For $\ell = 1$ and centered Gaussian w , $\Sigma^{(1)}$ is rotation-invariant with standard closed forms in terms of input cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$:

$$\Sigma^{(1)}(x, x') = \frac{\|x\| \|x'\|}{2\pi} ((\pi - \theta) \cos \theta + \sin \theta), \quad \theta = \arccos(\rho).$$

Given these definitions, we can now state the following composition result.

Theorem 2 (Gaussian process composition) *In the sequential infinite-width limit, the hidden states $\mathbf{h}^{(\ell)}$ form a composition of Gaussian processes:*

- *Conditionally on $\mathbf{h}^{(\ell-1)}$, for any finite collection of inputs $\{x_1, \dots, x_m\}$, the vector $(\mathbf{h}^{(\ell)}(x_1), \dots, \mathbf{h}^{(\ell)}(x_m))$ converges in distribution to a multivariate Gaussian.*
- *The conditional limiting Gaussian has mean zero and covariance matrix $[\Sigma^{(\ell)}(x_i, x_j)]_{i,j=1}^m$, where $\Sigma^{(\ell)}$ is the base kernel defined in Definition 1 and depends on $\mathbf{h}^{(\ell-1)}$*

Proof See Appendix A.2. ■

Composition of Gaussian processes We highlight the main difference between the base kernel and the NNGP kernel arising in classical NTK analysis : Since $\mathbf{h}^{(\ell-1)}$ itself is a (conditional) Gaussian process, the overall distribution is a composition of Gaussian processes, not a single Gaussian process.

3. NTK recursion

We now state the main result characterizing the NTK recursion for MMNNs. This recursion governs how the kernel evolves through layers and determines the training dynamics in the infinite-width limit.

Theorem 3 (Infinite-width NTK recursion) *Under NTK parameterization $1/\sqrt{n_\ell}$ and a sequential limit $n_1 \rightarrow \infty, \dots, n_L \rightarrow \infty$ with $\mathbf{w}^{(\ell)} \sim \mathcal{N}(0, \mathbf{I}_{d_{\ell-1}})$ i.i.d. frozen after initialization, the layer- ℓ NTK satisfies*

$$\Theta^{(0)}(x, x') = 0, \quad (4)$$

$$\Theta^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'), \quad (5)$$

$$\Theta^{(\ell)}(x, x') = 1 + \frac{1}{r} \Theta^{(\ell-1)}(x, x') \cdot \dot{\Sigma}^{(\ell)}(x, x') + \frac{1}{r} \Sigma^{(\ell)}(x, x'), \quad \ell \geq 2, \quad (6)$$

where the factor $1/r$ appears for bottleneck layers $\ell \geq 2$ (low-rank regime). For $\ell = 1$, $\Sigma^{(1)}$ is deterministic; for $\ell > 1$, $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are random fields whose fluctuations shrink with the rank r of bottlenecks.

Furthermore, the gradient flow dynamics in the NTK regime satisfy

$$\frac{d}{dt} f_t(x) = - \int \Theta^{(L)}(x, x') (f_t(x') - y(x')) d\mu(x'),$$

where μ is the data distribution and y is the target function.

Proof: See Appendix A.1, Proof A.1.

No weights particle coupling We can now interpret the structure of the recursion. Similarly to the original MLPs NTK recursion Jacot et al. (2018), the additive term $\Sigma^{(\ell)}$ encodes fresh basis contribution at layer ℓ , while the multiplicative coupling $\Theta^{(\ell-1)} \cdot \dot{\Sigma}^{(\ell)}$ reflects the backward through layers via the derivative kernel, since there is no coupling between weights particles between 2 layers during training due to random features in the middle. Nonetheless, this decoupling that makes the training easier to tackled under the mean-field limit is effectively lost at the NTK limit and allows to factorize the derivative kernel.

We can now derive an explicit closed-form expression for the NTK at any depth by expanding the recursion.

Corollary 4 (General L-layer NTK; closed form) *Under the assumptions of Theorem 3, for any $L \geq 1$, the NTK admits the explicit form:*

$$\Theta^{(L)}(x, x') = \sum_{\ell=1}^L \frac{1}{r^{L-\ell}} \left(\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^L \frac{1}{r^{\max(1, L-\ell)}} \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x'), \quad (7)$$

where the first sum has bias contributions (empty product = 1 when $\ell = L$) and the second has fresh-basis contributions; each bottleneck layer contributes a factor $1/r$. For $L = 2$: $\Theta^{(2)} = 1 + \frac{1}{r} (\dot{\Sigma}^{(2)} + \Sigma^{(1)} \dot{\Sigma}^{(2)} + \Sigma^{(2)})$.

Remark 5 *Equivalently, expanding the recursion yields a quadratic number of terms. For example, for $L = 3$:*

$$\begin{aligned} \Theta^{(3)}(x, x') = & 1 + \frac{1}{r} \dot{\Sigma}^{(2)} \cdot \dot{\Sigma}^{(3)} + \frac{1}{r} \dot{\Sigma}^{(3)} \\ & + \frac{1}{r^2} \Sigma^{(1)} \cdot \dot{\Sigma}^{(2)} \cdot \dot{\Sigma}^{(3)} + \frac{1}{r} \Sigma^{(2)} \cdot \dot{\Sigma}^{(3)} + \frac{1}{r} \Sigma^{(3)}. \end{aligned} \quad (8)$$

Each bottleneck layer contributes a factor $1/r$; layer 1 has no such factor. In total, there are $2L$ terms for depth L .

Proof By expanding the recursion in Theorem 3. See Appendix A.4 for the detailed proof by induction. $\blacksquare$

Structure at depth L . Iterating the recursion yields a quadratic number of terms. From the explicit form (7), we see two types of contributions:

- **Bias terms:** Each layer's $+1$ contribution multiplies by $1/r^{L-\ell}$ (for bottleneck layers) and through all subsequent derivative kernels, giving L terms.
- **Fresh basis terms:** Each layer's $\Sigma^{(\ell)}$ contribution carries $1/r^{L-\ell}$ and propagates through subsequent derivative kernels, giving L terms.

In total, the expansion contains $2L$ terms, each representing a different path through the network architecture. This linear growth in the number of terms (with respect to depth) becomes quadratic when considering all combinations of terms in finite-width corrections, making explicit computation and analysis very difficult to handle for deep networks.

3.1. Large-depth regime: probabilistic kernel and exponential decay

We now outline how to adapt the recursion for large depth L , focusing on the probabilistic (mean) kernel and the role of the $1/r$ scaling at low-rank bottlenecks.

Probabilistic kernel recursion and GP concatenation. For $\ell > 1$, $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are random fields because they depend on $\mathbf{h}^{(\ell-1)}$, which is the output of a (conditional) Gaussian process. By Theorem 2, the hidden states form a *composition of GPs*: each layer's output is conditionally Gaussian given the previous layer, but the unconditional law is a deep GP. The concatenation structure implies that at layer ℓ , the pair $(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$ follows a $2r$ -dimensional Gaussian (in the bottleneck regime), with covariance determined by $\Sigma^{(\ell-1)}$. The base kernel $\Sigma^{(\ell)}$ and derivative kernel $\dot{\Sigma}^{(\ell)}$ are thus expectations over $\mathbf{w}^{(\ell)}$, conditioned on this Gaussian pair; they inherit randomness from $\Sigma^{(\ell-1)}$.

The *mean* (probabilistic) NTK $\tilde{\Theta}^{(L)} = \mathbb{E}[\Theta^{(L)}]$ satisfies a recursion involving expectations of products:

$$\tilde{\Theta}^{(\ell)} = 1 + \mathbb{E}[\Theta^{(\ell-1)} \cdot \dot{\Sigma}^{(\ell)}] + \mathbb{E}[\Sigma^{(\ell)}], \quad \ell \geq 1. \quad (9)$$

Since $\Theta^{(\ell-1)}$ and $\Sigma^{(\ell)}, \dot{\Sigma}^{(\ell)}$ are not independent (they share the randomness of $\mathbf{h}^{(\ell-1)}$), we have $\mathbb{E}[\Theta^{(\ell-1)} \cdot \dot{\Sigma}^{(\ell)}] \neq \tilde{\Theta}^{(\ell-1)} \cdot \mathbb{E}[\dot{\Sigma}^{(\ell)}]$ in general. The correct expansion requires tracking covariances

between terms across layers. Under Fisher–Kibble decoupling at each layer, the randomness decomposes into independent angular and radial parts, which can be handled iteratively; the technical difficulty lies in controlling the nested expectations.

1/r scaling at each bottleneck. In the low-rank regime, the derivative kernel at layer $\ell \geq 2$ scales with $1/r$. For ReLU and isotropic weights with $\text{Cov}(w) = I_r/r$ on the r -dimensional bottleneck output, one has

$$\dot{\Sigma}^{(\ell)}(x, x') = \frac{1}{r} \mathbb{E}_w \left[\dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x)) \dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x')) \|w\|^2 \right] = \frac{1}{r} \bar{\Sigma}^{(\ell)}(\rho_\ell),$$

where $\rho_\ell = \cos \angle(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$ and $\bar{\Sigma}^{(\ell)}(\rho)$ denotes the corresponding scalar derivative kernel evaluated at cosine similarity $\rho \in [-1, 1]$. For ReLU under the EOC normalization $\text{Tr}(\text{Cov}(w)) = 1$, the derivative kernel has the explicit form

$$\bar{\Sigma}^{(\ell)}(\rho) = \mathbb{P}(Z > 0, Z' > 0) = \frac{1}{2} - \frac{\arccos(\rho)}{2\pi} \in \left[0, \frac{1}{2}\right],$$

where (Z, Z') is centered Gaussian with correlation ρ . In particular, $\sup_{\rho \in [-1, 1]} \bar{\Sigma}^{(\ell)}(\rho) = 1/2$, uniformly in ℓ . Similarly, $\Sigma^{(\ell)}$ involves a norm product $\|\mathbf{h}^{(\ell-1)}\|^2/r$ with $\mathbb{E}[\cdot] \rightarrow 1$ and $\text{Var}(\cdot) = O(1/r)$. Thus, at each bottleneck layer, both the multiplicative and additive contributions carry a $1/r$ factor relative to the full-width case.

Exponential decay with depth. Assume for simplicity that all layers $\ell \geq 2$ are bottlenecks with the same rank r . For ReLU under EOC we may take the explicit uniform bound

$$\bar{\Sigma}^{(\ell)}(\rho) \leq c_0 := \frac{1}{2}, \quad \forall \rho \in [-1, 1], \forall \ell \geq 2.$$

From the closed form (7), the contribution from layer ℓ to $\Theta^{(L)}$ is multiplied by $\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}$. With $\dot{\Sigma}^{(k)} = \bar{\Sigma}^{(k)}(\rho_k)/r$, we obtain

$$\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)} \sim \frac{1}{r^{L-\ell}} \prod_{k=\ell+1}^L \bar{\Sigma}^{(k)}(\rho_k) \leq \frac{c_0^{L-\ell}}{r^{L-\ell}}.$$

Hence, contributions from early layers ($\ell \ll L$) decay *exponentially* in $L - \ell$ when $c_0/r < 1$. The dominant terms in $\Theta^{(L)}$ come from the last $O(1)$ layers; the NTK becomes increasingly localized to the top of the network as depth grows.

Proposition 6 (Informal: depth-induced decay) *Under the assumptions above, for r fixed and $L \rightarrow \infty$, early-layer contributions to $\Theta^{(L)}$ are suppressed by a factor $(c/r)^{L-\ell}$ for some $c = O(1)$. In particular, when $c/r < 1$, the NTK is dominated by contributions from the last $O(\log r)$ layers. The mean NTK $\tilde{\Theta}^{(L)}$ concentrates around a limit that depends primarily on these top layers.*

Open directions. A full treatment requires: (i) a rigorous recursion for $\tilde{\Theta}^{(\ell)}$ with explicit covariance bounds; (ii) verification that $\bar{\Sigma}^{(\ell)}$ remains uniformly bounded in ℓ and ρ_ℓ ; (iii) extension of Fisher–Kibble decoupling to nested layers to control $\mathbb{E}[\Theta^{(\ell-1)} \dot{\Sigma}^{(\ell)}]$. The exponential decay suggests that very deep MMNNs may exhibit a “effective depth” phenomenon similar to ResNets, where only the top layers meaningfully contribute to the NTK.

We now specialize to the three-layer case, where we can obtain a particularly compact form under a homogeneity assumption on the activation function.

Corollary 7 (three-layer NTK and homogeneous activation compact form) *Under the assumptions of Theorem 3 with $L = 2$,*

$$\begin{aligned}\Theta^{(1)}(x, x') &= 1 + \Sigma^{(1)}(x, x'), \\ \Theta^{(2)}(x, x') &= 1 + \frac{1}{r} \Theta^{(1)}(x, x') \cdot \dot{\Sigma}^{(2)}(x, x') + \frac{1}{r} \Sigma^{(2)}(x, x').\end{aligned}$$

Proof Immediate from Theorem 3. ■

We can now state a more compact representation under a homogeneity assumption.

3.2. Microscopic distributions: Fisher–Kibble decoupling

Lemma 8 (Fisher–Kibble decoupling) *Let x, y be input vectors and let x_1, y_1 denote their rank- r random projections. Define the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ and squared norms $u = \|x_1\|^2, v = \|y_1\|^2$. Then:*

- ρ_1 follows Fisher’s correlation distribution [Fisher \(1915\)](#); [Hotelling \(1953\)](#), centered at the true correlation ρ .
- (u, v) follow Kibble’s (1941) bivariate Gamma (chi-square) law.
- Angular and radial parts are independent: $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) p_{\text{Kibble}}(u, v)$.

Proof: See Appendix B.1, Proof B.1.

Homogeneity assumption enabling decoupling. The independence of angular and radial components relies on the *positive 1-homogeneity* of the ReLU activation: $\sigma(\alpha z) = \alpha \sigma(z)$ for $\alpha > 0$. This property ensures that the base kernel factorizes as

$$\Sigma^{(1)}(x, x') = \|x\| \|x'\| \cdot \tilde{\Sigma}^{(1)}(\rho),$$

where $\tilde{\Sigma}^{(1)}(\rho)$ depends only on the cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$, enabling the separation of radial norms from angular correlations. For isotropic Gaussian projections, rotational invariance guarantees that the empirical correlation ρ_1 and the norms $(\|x_1\|, \|y_1\|)$ are statistically independent, yielding the factorization stated in the lemma.

This decoupling result enables us to derive the asymptotic distribution of the empirical kernel. We can state the following central limit theorem.

In the three-layer low-rank setting, denote

$$\rho_1 = \cos \angle(\mathbf{h}^{(1)}(x), \mathbf{h}^{(1)}(x')), \quad w_r = \frac{\|\mathbf{h}^{(1)}(x)\| \|\mathbf{h}^{(1)}(x')\|}{r},$$

then both ρ_1 and w_r are random variables induced by the randomness of the features and the rank- r projection; their values fluctuate with the initialization and with r . The NTK admits the compact representation

$$\Theta^{(2)}(x, y) = 1 + \frac{1}{r} \Theta^{(1)}(\rho_1) \left(1 - \frac{\arccos(\rho_1)}{\pi} \right) + \frac{1}{r} w_r \Sigma^{(1)}(\rho_1)$$

with $\Sigma^{(1)}(u) = \frac{1}{\pi} (\sqrt{1-u^2} + u(1 - \arccos u))$.

Remark (Fisher–Kibble decoupling and independence). We can now characterize the statistical properties of the empirical correlation and norms. By rotational invariance, the empirical correlation and squared norms follow Fisher–Kibble decoupling [Fisher \(1915\)](#); [Hotelling \(1953\)](#):

$$\rho_1 \sim \text{Fisher}(\rho, r), \quad (\|\mathbf{h}^{(1)}(x)\|^2, \|\mathbf{h}^{(1)}(x')\|^2) \sim \text{Kibble}(r, \rho),$$

We now state the explicit forms of these distributions. The **Fisher distribution density** [Fisher \(1915\)](#) is

$$p(\rho_1 \mid \rho, r) = \frac{(r-2) \Gamma(r-1) (1-\rho^2)^{\frac{r-1}{2}} (1-\rho_1^2)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1-\rho\rho_1)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho\rho_1}{2}\right),$$

for $r > 2$. For large r , the inverse Gudermann function applied to the complementary angle follows a normal distribution:

$$\text{gd}^{-1}\left(\frac{\pi}{2} - \arccos(\rho_1)\right) \approx \mathcal{N}\left(\text{arctanh}(\rho), \frac{1}{r-3}\right)$$

where $\rho_1 = \cos(\theta_1)$ is the empirical correlation and θ_1 is the angle between the projected vectors. In particular, this implies $\text{Var}(\rho_1) = O(1/r)$.

Similarly, **Kibble’s Kibble (1941) bivariate chi-square law** for $(u, v) = (\|\mathbf{h}^{(1)}(x)\|^2, \|\mathbf{h}^{(1)}(x')\|^2)$ has density

$$f(u, v) = \frac{(uv)^{\frac{r/2-1}{2}} \exp\left(-\frac{u+v}{2(1-\rho^2)}\right)}{\Gamma(r/2) (2(1-\rho^2))^{\frac{r}{2}+1} \rho^{\frac{r/2-1}{2}}} I_{\frac{r}{2}-1}\left(\frac{\rho\sqrt{uv}}{1-\rho^2}\right), \quad u, v \geq 0,$$

where I_ν is the modified Bessel function of the first kind. Defining $w_r = \sqrt{uv}/r$, concentration results (see [Theorem 22](#)) imply $\mathbb{E}[w_r] \rightarrow 1$ and $\text{Var}(w_r) = O(1/r)$.

The moments of this distribution are given by:

$$\mathbb{E}[u] = \mathbb{E}[v] = r, \quad \text{Var}(u) = \text{Var}(v) = 2r, \quad \text{Cov}(u, v) = 2r \rho^2.$$

The derivation of these distributions from Gaussian projections and the independence property are given in [Appendix B.1](#).

4. Depth scaling of the RF-LR NTK

We adapt the depth-scaling viewpoint of [Terjék and González-Sánchez \(2025\)](#) (inverse-cosine-distance propagation and its spectral consequences) to our doubly random RF-LR kernel. A crucial caveat is that the explicit closed form for the *MLP* NTK at EOC does *not* directly apply to RF-LR: RF-LR trains only the low-rank readouts $A^{(\ell)}$ with frozen random features $w^{(\ell)}$, hence the NTK is governed by our recursion (Theorem 3) rather than the full-MLP formula. What does carry over is the geometry of *activation correlations* and the associated inverse cosine distance propagation, which we use to control $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ across depth.

4.1. Correlation propagation and inverse cosine distances

For ReLU activation, the RF-LR architecture uses the same EOC parameterization as standard MLPs [Hayou et al. \(2018\)](#). The forward correlation map induced by ReLU is the same as in [Terjék and González-Sánchez \(2025\)](#); in particular, if we define a deterministic cosine recursion $\rho_k = \varrho(\rho_{k-1})$ (the infinite-width/full-rank idealization), then for ReLU $\Delta_\phi = 1/2$ (i.e. $a = b = 1/\sqrt{2}$ in (a, b) -ReLU notation).

Definition 9 (Cosine map for ReLU at EOC) *For ReLU, set $\Delta_\phi = 1/2$. The cosine map $\varrho : [-1, 1] \rightarrow [-1, 1]$ and its derivative ϱ' satisfy*

$$\varrho(\rho) = \rho + \Delta_\phi \frac{2}{\pi} (\sqrt{1 - \rho^2} - \rho \arccos(\rho)), \quad \varrho'(\rho) = 1 - \Delta_\phi \frac{2}{\pi} \arccos(\rho).$$

We write $\rho_k = \varrho(\rho_{k-1})$ for the (deterministic) limiting correlation at layer k .

To a cosine $\rho \in (-1, 1)$, we associate the squared cosine distance $z = (1 - \rho)/2 \in (0, 1)$ and the inverse cosine distance $w = z^{-1/2} \in (1, \infty)$.

Definition 10 (Squared and inverse cosine distance maps) *Define $\zeta : [0, 1] \rightarrow [0, 1]$ by $\zeta(z) = (1 - \varrho(1 - 2z))/2$ and $\omega : (1, \infty) \rightarrow (1, \infty)$ by $\omega(w) = \zeta(w^{-2})^{-1/2}$. Then ω is convex and satisfies*

$$\omega(w) = w + \Delta_\phi \frac{4}{3\pi} + \frac{3}{2} \left(\Delta_\phi \frac{4}{3\pi} \right)^2 w^{-1} + O(w^{-2}).$$

Proposition 11 (Inverse cosine distance propagation for RF-LR) *Given $w \in (1, \infty)$, for $k \in \mathbb{N}$ we have*

$$|\omega^{\circ k}(w) - (w + \Delta_\phi \frac{4}{3\pi}(k-1) + \Delta_\phi \frac{2}{\pi} \log(\Delta_\phi^{-1} \frac{3\pi}{4} w + k - 1))| \leq O(1).$$

Proof This is a statement about the iterates of the inverse cosine distance map ω , so what we must justify is that ω is indeed the correct map governing the *RF-LR* correlation recursion.

Step 1: RF-LR correlation map (idealized) equals the ReLU cosine map. Fix a layer $\ell \geq 2$ and two inputs (x, x') . Write the RF-LR layer as

$$\mathbf{h}^{(\ell)}(x) = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \boldsymbol{\sigma}(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x)), \quad \mathbf{h}^{(\ell)}(x') = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \boldsymbol{\sigma}(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x')),$$

where $A_j^{(\ell)} \in \mathbb{R}^r$ is the (random) readout column and $w_j^{(\ell)}$ is the frozen random feature direction. Assume the standard RF-LR initialization: $\{A_j^{(\ell)}\}_{j=1}^{n_\ell}$ are i.i.d. centered with $\mathbb{E}[A_j^{(\ell)} A_j^{(\ell)\top}] = \sigma_A^2 I_r$, independent of $\{w_j^{(\ell)}\}_{j=1}^{n_\ell}$, and take the sequential infinite-width limit $n_\ell \rightarrow \infty$ (Definition 1).

Condition on $\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x')$. By independence across j ,

$$\mathbb{E}[\mathbf{h}^{(\ell)}(x) \cdot \mathbf{h}^{(\ell)}(x') \mid \mathbf{h}^{(\ell-1)}] = \frac{1}{n_\ell} \sum_{j=1}^{n_\ell} \mathbb{E}[\|A_j^{(\ell)}\|^2] \mathbb{E}[\sigma(U_j) \sigma(V_j) \mid \mathbf{h}^{(\ell-1)}],$$

where $(U_j, V_j) = (w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x), w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x'))$ is a centered bivariate Gaussian conditional on $\mathbf{h}^{(\ell-1)}$, with correlation equal to the (random) sample cosine $\rho_{\ell-1}(x, x')$. The same computation gives

$$\mathbb{E}[\|\mathbf{h}^{(\ell)}(x)\|^2 \mid \mathbf{h}^{(\ell-1)}] = \frac{1}{n_\ell} \sum_{j=1}^{n_\ell} \mathbb{E}[\|A_j^{(\ell)}\|^2] \mathbb{E}[\sigma(U_j)^2 \mid \mathbf{h}^{(\ell-1)}],$$

and similarly for x' . Passing to the limit $n_\ell \rightarrow \infty$, the empirical averages converge to the corresponding ReLU Gaussian expectations; crucially, the (isotropic) factor $\mathbb{E}\|A_j^{(\ell)}\|^2$ cancels in the cosine ratio. Therefore, in the *deterministic mean-field idealization* (where one replaces the random sample cosines $\rho_{\ell-1}(x, x')$ by their deterministic correlation coefficients and studies the induced map on those coefficients), the RF-LR correlation recursion is governed by the same ReLU cosine map ϱ as in [Terjék and González-Sánchez \(2025\)](#). For finite bottleneck dimension r , $\rho_{\ell-1}(x, x')$ remains random; justifying the deterministic recursion requires r -dependent concentration (which in this work is handled via Fisher–Kibble/Fisher-type asymptotics and leads to $O(r^{-1/2})$ -scale fluctuations).

Step 2: reduce to iterates of ω . With $z_k = (1 - \rho_k)/2$ and $w_k = z_k^{-1/2}$, we have $z_k = \zeta(z_{k-1})$ and therefore $w_k = \omega(w_{k-1})$, i.e. $w_k = \omega^{\circ(k-1)}(w_1)$.

Step 3: asymptotics of $\omega^{\circ k}$. Since ω is exactly the same inverse cosine distance map as in [Terjék and González-Sánchez \(2025\)](#) (it is defined from the ReLU cosine map ϱ), the claimed bound is precisely the iterate asymptotic proved in ([Terjék and González-Sánchez, 2025](#), Proposition 3). ■

4.2. RF-LR NTK: recursion-driven depth scaling (not the full-MLP closed form)

For full-width MLPs at EOC, the limiting NTK admits an explicit expression in terms of the cosine map and its derivative ([Terjék and González-Sánchez, 2025](#), Proposition 4). This identity relies on the fact that *all* weight layers are trained. In contrast, RF-LR trains only the low-rank readouts $A^{(\ell)}$ and freezes $w^{(\ell)}$; therefore the RF-LR NTK is given by our recursion (Theorem 3) and its closed form (Corollary 4, Eq. (7)) in terms of $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$.

The Terjék machinery is still useful here, but at the level of controlling the depth-evolution of correlations (hence of $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$), not by importing the full-MLP NTK identity.

Proposition 12 (Geometric suppression of early layers under bottlenecks) *Assume $r > 1$, and for all $\ell \geq 2$ and all input pairs (x, x') we have the bounds $0 \leq \dot{\Sigma}^{(\ell)}(x, x') \leq 1$ and $|\Sigma^{(\ell)}(x, x')| \leq$*

$C \|x\| \|x'\|$ for some constant $C > 0$. Then the RF-LR NTK entries satisfy the uniform depth bound

$$|\Theta^{(L)}(x, x')| \leq \sum_{\ell=1}^L \frac{1}{r^{L-\ell}} + C \|x\| \|x'\| \sum_{\ell=1}^L \frac{1}{r^{\max(1, L-\ell)}} \leq \frac{r}{r-1} + \frac{C \|x\| \|x'\|}{r-1},$$

and each layer- ℓ contribution in Eq. (7) is suppressed by the factor $r^{-(L-\ell)}$ times the product of subsequent $\Sigma^{(k)}$ terms.

Remark 13 (Main qualitative difference to full-width EOC NTK) In [Terjék and González-Sánchez \(2025\)](#), the full-width MLP EOC NTK has $\lambda_{\max}$ growing on the order of L . Proposition 12 shows that, in RF-LR, the explicit $1/r$ bottleneck scaling forces a geometric depth suppression: for fixed $r > 1$, the RF-LR NTK entries (hence operator norms) do not grow linearly with depth.

4.3. Inverse cosine distance matrices and spectral bounds

To formulate dataset-level depth scaling, we work with the deterministic correlation recursion (the mean/infinite-width idealization). Given a dataset $\{x_1, \dots, x_n\} \subset \mathbb{R}^d$, set

$$\rho_{1,ij} = \frac{\langle x_i, x_j \rangle}{\|x_i\| \|x_j\|} \in [-1, 1], \quad \rho_{k,ij} = \varrho(\rho_{k-1,ij}), \quad k \geq 2.$$

Define the layer- k inverse cosine distance matrix $W_k \in \mathbb{S}^n$ by

$$(W_k)_{i,i} = 0, \quad (W_k)_{i_1, i_2} = \left(\frac{1 - \rho_{k, i_1 i_2}}{2} \right)^{-1/2}, \quad i_1 \neq i_2,$$

so that W_k is the inverse cosine distance matrix associated to ρ_k .

Proposition 14 (Spectral bounds for RF-LR inverse cosine distance matrices) Assume EOC initialization and a dataset with no parallel pairs. For each $k \in [1 : L]$ there exists $\widehat{W}_k \in (1, \infty)$ with $\widehat{W}_k = \Theta_{W_1}(1)$ such that

$$\|W_k - \omega^{\circ(k-1)}(\widehat{W}_k) (\mathbf{1}_n \mathbf{1}_n^\top - I_n)\| \leq O(\Delta_\phi n^{-1} k) + O(1).$$

Proof This is exactly ([Terjék and González-Sánchez, 2025](#), Proposition 6) applied to the deterministic recursion $\rho_{k,ij} = \varrho(\rho_{k-1,ij})$. The proof uses only convexity/monotonicity of ω and matrix comparison arguments. $\blacksquare$

4.4. Spectral consequences (depth scaling)

Proposition 12 implies that, for fixed $r > 1$, the depth-dependence of the RF-LR NTK is governed by a geometric series rather than a linear-in- L growth. Consequently, any operator-norm bound for the RF-LR Gram matrix $\mathbf{K}^{(L)}$ obtained from entrywise control and standard matrix inequalities will be *uniform* in L , in sharp contrast with the full-width EOC NTK of [Terjék and González-Sánchez \(2025\)](#).

Deriving Terjék-tight eigenvalue and condition-number bounds for $\mathbf{K}^{(L)}$ in RF-LR requires combining: (i) the inverse-cosine-distance control of correlations (Propositions 11 and 14); with (ii) the RF-LR NTK expansion (7) that introduces the $1/r$ bottleneck weights.

4.5. A Terjék-style depth theorem for the RF-LR NTK (new kernel, new scaling)

We now give an explicit depth scaling statement for the *mean* RF-LR NTK that is not a copy of [Terjék and González-Sánchez \(2025\)](#): the RF-LR recursion contains an explicit $1/r$ bottleneck factor, so the kernel magnitude *saturates* in depth, while the diagonal–off-diagonal gap decays with depth in the mean recursion.

Scalar ReLU kernels as functions of cosine similarity. For unit-variance ReLU features, write $\theta = \arccos(\rho)$. The normalized base and derivative kernels are

$$s(\rho) = \frac{1}{2\pi}((\pi - \theta) \cos \theta + \sin \theta), \quad \dot{s}(\rho) = \frac{1}{2} - \frac{\theta}{2\pi}.$$

Near $\rho = 1$, parameterize by the inverse cosine distance $w = ((1 - \rho)/2)^{-1/2}$, so that $\rho = 1 - 2w^{-2}$ and $\theta = 2w^{-1} + O(w^{-3})$. Then

$$\dot{s}(\rho) = \frac{1}{2} - \frac{1}{\pi w} + O(w^{-3}), \quad s(\rho) = \frac{1}{2} - \frac{1}{w^2} + O(w^{-3}). \quad (10)$$

Deterministic proxy (mean-field) RF-LR entrywise recursion. Fix an input pair (x, x') with input cosine similarity $\rho_1 \in (-1, 1)$. Define a *deterministic* correlation recursion $\rho_k = \varrho(\rho_{k-1})$ and $w_k = ((1 - \rho_k)/2)^{-1/2}$. Define the scalar deterministic proxy RF-LR NTK by

$$\Theta^{(1)}(\rho_1) = 1 + s(\rho_1), \quad \Theta^{(k)}(\rho_1) = 1 + \frac{1}{r} \Theta^{(k-1)}(\rho_1) \dot{s}(\rho_k) + \frac{1}{r} s(\rho_k), \quad k \geq 2. \quad (11)$$

This recursion should be viewed as a *mean-field proxy*: it is obtained from Theorem 3 by replacing the random fields $\Sigma^{(k)}, \dot{\Sigma}^{(k)}$ with their scalar ReLU dual functions $s(\cdot), \dot{s}(\cdot)$ evaluated along the deterministic correlation coefficients ρ_k . Importantly, for finite bottleneck dimension r , the *sample* cosine similarity $\langle \mathbf{h}^{(k)}(x), \mathbf{h}^{(k)}(x') \rangle / (\|\mathbf{h}^{(k)}(x)\| \|\mathbf{h}^{(k)}(x')\|)$ is random; (11) does not assert that this random cosine equals ρ_k almost surely, nor that $\Theta^{(k)}$ equals the exact annealed mean $\mathbb{E}[\Theta^{(k)}(x, x')]$ of the full doubly-random deep GP. Rather, ρ_k is the deterministic correlation *parameter* governing the covariance of the limiting (conditional) Gaussian pair at depth k , and the usefulness of (11) rests on r -concentration: for a jointly Gaussian pair in $\mathbb{R}^r$ with correlation parameter ρ_k , the induced sample cosine concentrates around ρ_k with variance $O(1/r)$ (equivalently, $\mathbb{E}|\hat{\rho}_k - \rho_k| \lesssim r^{-1/2}$).

Theorem 15 (Depth scaling for the mean RF-LR NTK) Fix $r > 1$ and $\rho_1 \in (-1, 1)$. Let $\Theta^{(k)}(\rho_1)$ be defined by (11). Then:

- **(Correlation alignment)** There exist $k_0 \in \mathbb{N}$ and constants $0 < c_- < c_+ < \infty$ (depending only on ρ_1) such that for all $k \geq k_0$,

$$c_- k \leq w_k \leq c_+ k, \quad \text{hence} \quad 1 - \rho_k = 2w_k^{-2} = O(k^{-2}).$$

- **(Kernel saturation)** $\Theta^{(k)}(\rho_1) \rightarrow \Theta_\star(r)$ as $k \rightarrow \infty$, where $\Theta_\star(r)$ is the fixed point of the limiting recursion obtained by setting $\dot{s} = s = 1/2$, namely

$$\Theta_\star(r) = \frac{2r + 1}{2r - 1}.$$

Moreover, there exists $C_\Theta < \infty$ (depending on r, ρ_1) such that for all $k \geq 1$,

$$|\Theta^{(k)}(\rho_1) - \Theta_\star(r)| \leq \frac{C_\Theta}{k}.$$

- **(Depth-induced gap decay)** Let $\Theta_{\text{diag}}^{(k)} = \Theta^{(k)}(1)$ and $\Theta_{\text{off}}^{(k)} = \Theta^{(k)}(\rho_1)$. Then

$$\Theta_{\text{diag}}^{(k)} - \Theta_{\text{off}}^{(k)} = \Theta\left(\frac{1}{r k}\right),$$

i.e. there exist $k_1 \in \mathbb{N}$ and constants $0 < c_\Delta < C_\Delta < \infty$ (depending on r, ρ_1) such that for all $k \geq k_1$,

$$\frac{c_\Delta}{r k} \leq \Theta_{\text{diag}}^{(k)} - \Theta_{\text{off}}^{(k)} \leq \frac{C_\Delta}{r k}.$$

Proof

Step 1: correlation alignment. By Proposition 11, there exists a constant $C_w < \infty$ such that for all $k \geq 1$,

$$w_k = \omega^{\circ(k-1)}(w_1) = w_1 + c_0(k-1) + c_1 \log(c_2 w_1 + k - 1) + \varepsilon_k, \quad |\varepsilon_k| \leq C_w,$$

where $c_0 = \Delta_\phi \frac{4}{3\pi} > 0$, $c_1 = \Delta_\phi \frac{2}{\pi}$, $c_2 = \Delta_\phi^{-1} \frac{3\pi}{4}$. In particular, $w_k \geq w_1 + c_0(k-1) - C_w$, so choosing k_0 large enough gives $w_k \geq \frac{c_0}{2} k$ for all $k \geq k_0$. Similarly $w_k \leq c_+ k$ for some $c_+ < \infty$ and all $k \geq k_0$. Thus $c_- k \leq w_k \leq c_+ k$ for $k \geq k_0$, and $1 - \rho_k = 2w_k^{-2} = O(k^{-2})$.

Step 2: kernel saturation. Write $\Theta_k = \Theta^{(k)}(\rho_1)$ and rewrite (11) as

$$\Theta_k = b_k + a_k \Theta_{k-1}, \quad a_k = \frac{1}{r} \dot{s}(\rho_k), \quad b_k = 1 + \frac{1}{r} s(\rho_k).$$

For the limiting constant-coefficient recursion (corresponding to $\rho_k \rightarrow 1$), we have $a = \frac{1}{r} \dot{s}(1) = \frac{1}{2r}$, $b = 1 + \frac{1}{r} s(1) = 1 + \frac{1}{2r}$, hence $\Theta_\star(r) = b/(1-a) = (2r+1)/(2r-1)$.

Set $e_k = \Theta_k - \Theta_\star(r)$. Then

$$e_k = a_k e_{k-1} + (b_k - b) + (a_k - a) \Theta_\star(r).$$

Using (10) and $w_k \geq c_- k$ for $k \geq k_0$, there exists $C < \infty$ such that for all $k \geq k_0$,

$$|a_k - a| = \frac{1}{r} \left| \dot{s}(\rho_k) - \frac{1}{2} \right| \leq \frac{C}{r} \frac{1}{w_k} \leq \frac{C}{r} \frac{1}{k}, \quad |b_k - b| = \frac{1}{r} \left| s(\rho_k) - \frac{1}{2} \right| \leq \frac{C}{r} \frac{1}{w_k^2} \leq \frac{C}{r} \frac{1}{k^2}.$$

Moreover $0 \leq a_k \leq a = 1/(2r) < 1$. Hence for all $k \geq k_0$,

$$|e_k| \leq a |e_{k-1}| + \frac{C'}{k}$$

for some $C' < \infty$. A standard induction (discrete Grönwall) yields $|e_k| \leq C_\Theta/k$ for all $k \geq 1$, in particular $e_k \rightarrow 0$ and $\Theta_k \rightarrow \Theta_\star(r)$.

To make the $O(1/k)$ rate explicit, fix $k \geq k_0$ and iterate the one-step inequality:

$$|e_k| \leq a^{k-k_0} |e_{k_0}| + \sum_{j=k_0+1}^k a^{k-j} \frac{C'}{j}.$$

Since $j \geq k_0 + 1$ implies $1/j \leq 1/k_0$ and also $j \leq k$ implies $1/j \leq 1/k$, we can bound

$$\sum_{j=k_0+1}^k a^{k-j} \frac{1}{j} \leq \frac{1}{k} \sum_{t=0}^{k-k_0-1} a^t \leq \frac{1}{k} \cdot \frac{1}{1-a}.$$

Also $a^{k-k_0}|e_{k_0}| \leq |e_{k_0}| \leq (k_0|e_{k_0}|)/k$. Therefore for all $k \geq k_0$,

$$|e_k| \leq \frac{1}{k} \left(k_0|e_{k_0}| + \frac{C'}{1-a} \right).$$

Absorbing the finitely many values $k < k_0$ into the constant gives $|e_k| \leq C_\Theta/k$ for all $k \geq 1$, as claimed.

Step 3: diagonal–off-diagonal gap. Let $\Theta_k^{\text{diag}} = \Theta^{(k)}(1)$ and $\Theta_k^{\text{off}} = \Theta^{(k)}(\rho_1)$, and set $\Delta_k = \Theta_k^{\text{diag}} - \Theta_k^{\text{off}} \geq 0$. The diagonal recursion is $\Theta_k^{\text{diag}} = b + a \Theta_{k-1}^{\text{diag}}$, while the off-diagonal recursion is $\Theta_k^{\text{off}} = b_k + a_k \Theta_{k-1}^{\text{off}}$. Subtracting gives the exact identity

$$\Delta_k = a \Delta_{k-1} + (a - a_k) \Theta_{k-1}^{\text{off}} + (b - b_k).$$

For k large, $\Theta_{k-1}^{\text{off}}$ is bounded above and below by positive constants (since $\Theta_k^{\text{off}} \rightarrow \Theta_\star(r) \in (1, \infty)$). Also by (10) and $w_k \asymp k$,

$$a - a_k = \frac{1}{r} \left(\frac{1}{2} - \dot{s}(\rho_k) \right) = \Theta \left(\frac{1}{r w_k} \right) = \Theta \left(\frac{1}{r k} \right), \quad b - b_k = \frac{1}{r} \left(\frac{1}{2} - s(\rho_k) \right) = O \left(\frac{1}{r w_k^2} \right) = O \left(\frac{1}{r k^2} \right).$$

Thus there exist k_1 and constants $0 < c < C < \infty$ such that for all $k \geq k_1$,

$$a \Delta_{k-1} + \frac{c}{r k} \leq \Delta_k \leq a \Delta_{k-1} + \frac{C}{r k}.$$

Iterating these comparison recursions yields $c_\Delta/(rk) \leq \Delta_k \leq C_\Delta/(rk)$ for all $k \geq k_1$, completing the proof. $\blacksquare$

Proposition 16 (Centered eigenvalue in an equicorrelated model) Fix $L \geq 2$ and $r > 1$. Consider a dataset $\{x_i\}_{i=1}^n$ with $\|x_i\| = 1$ and $\rho_{1,ij} = \rho_0 \in (-1, 1)$ for all $i \neq j$. Let

$$K_{ij}^{(L)} := \Theta^{(L)}(\rho_{1,ij}), \quad H := I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top, \quad K_c^{(L)} := H K^{(L)} H.$$

Then $K^{(L)}$ has the explicit form $K^{(L)} = \Theta^{(L)}(1) I_n + \Theta^{(L)}(\rho_0) (\mathbf{1}\mathbf{1}^\top - I_n)$. In particular, the nonzero eigenvalues of $K_c^{(L)}$ are all equal to

$$\lambda_\perp(K^{(L)}) = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_0),$$

with multiplicity $n - 1$. Moreover, by Theorem 15, there exist constants $0 < c_\Delta < C_\Delta < \infty$ (depending on r, ρ_0) such that for all sufficiently large L ,

$$\frac{c_\Delta}{r L} \leq \lambda_\perp(K^{(L)}) \leq \frac{C_\Delta}{r L}.$$

Proof The explicit form of $K^{(L)}$ follows from $\rho_{1,ii} = 1$ and $\rho_{1,ij} = \rho_0$ for $i \neq j$. The matrix $\mathbf{1}\mathbf{1}^\top - I_n$ has eigenvalue $n - 1$ along $\mathbf{1}$ and eigenvalue -1 on $\mathbf{1}^\perp$. Therefore $K^{(L)}$ has eigenvalue $\Theta^{(L)}(1) + (n - 1)\Theta^{(L)}(\rho_0)$ along $\mathbf{1}$ and eigenvalue $\Theta^{(L)}(1) - \Theta^{(L)}(\rho_0)$ on $\mathbf{1}^\perp$. Since H is the orthogonal projection onto $\mathbf{1}^\perp$, the nonzero eigenvalues of $K_c^{(L)} = H K^{(L)} H$ are exactly the $\mathbf{1}^\perp$ eigenvalues of $K^{(L)}$, with multiplicity $n - 1$. The $1/(rL)$ bounds then follow from Theorem 15 applied to $\rho_1 = \rho_0$. $\blacksquare$

Proposition 17 (Uncentered spike and bulk scaling in the equicorrelated model) *Under the assumptions of Proposition 16, the eigenvalues of the uncentered Gram matrix $K^{(L)}$ are explicit:*

$$\lambda_1(K^{(L)}) = \Theta^{(L)}(1) + (n-1)\Theta^{(L)}(\rho_0), \quad \lambda_2(K^{(L)}) = \dots = \lambda_n(K^{(L)}) = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_0).$$

In particular, $\lambda_1(K^{(L)})$ is a rank-one “spike” aligned with $\mathbf{1}$, while the spectrum on $\mathbf{1}^\perp$ equals $\lambda_\perp(K^{(L)})$.

Moreover, as $L \rightarrow \infty$ (with r fixed) the spike saturates:

$$\lambda_1(K^{(L)}) = n\Theta_\star(r) + O\left(\frac{n}{L}\right), \quad \Theta_\star(r) = \frac{2r+1}{2r-1} = 1 + O\left(\frac{1}{r}\right),$$

and the $\mathbf{1}^\perp$ eigenvalue satisfies $\lambda_\perp(K^{(L)}) = \Theta(1/(rL))$ by Proposition 16. Thus, in this model the depth/rank scaling is:

$$\lambda_{\max}(K^{(L)}) \asymp n\left(1 + \frac{1}{r}\right), \quad \lambda_{\min}(HK^{(L)}H) \asymp \frac{1}{rL}.$$

Proof The eigenvalue identities are the $\mathbf{1}$ and $\mathbf{1}^\perp$ eigenvalues established in the proof of Proposition 16. The spike saturation follows from Theorem 15: both $\Theta^{(L)}(1)$ and $\Theta^{(L)}(\rho_0)$ converge to $\Theta_\star(r)$ at rate $O(1/L)$, hence

$$|\lambda_1(K^{(L)}) - n\Theta_\star(r)| \leq |\Theta^{(L)}(1) - \Theta_\star(r)| + (n-1)|\Theta^{(L)}(\rho_0) - \Theta_\star(r)| = O\left(\frac{n}{L}\right).$$

Finally, $\lambda_{\max}(K^{(L)}) \asymp n(1 + 1/r)$ follows from $\lambda_1(K^{(L)}) \sim n\Theta_\star(r)$ and $\Theta_\star(r) = (2r+1)/(2r-1) = 1 + O(1/r)$, while $\lambda_{\min}(HK^{(L)}H) = \lambda_\perp(K^{(L)}) \asymp 1/(rL)$ follows from Proposition 16. ■

Remark 18 (Conditioning: uncentered vs centered) *The scaling $\lambda_{\max}(K^{(L)}) \asymp n(1 + 1/r)$ together with $\lambda_{\min}(HK^{(L)}H) \asymp 1/(rL)$ should not be read as a statement that “the condition number worsens like nrL ”: these quantities live on different invariant subspaces. For mean-zero targets (or after centering), the relevant operator is $HK^{(L)}H$ on $\mathbf{1}^\perp$. In the equicorrelated model, $HK^{(L)}H$ has a single nonzero eigenvalue $\lambda_\perp(K^{(L)})$ with multiplicity $n-1$, hence its condition number on $\mathbf{1}^\perp$ equals 1 for every L, r . What does deteriorate with L and r here is the scale $\lambda_\perp(K^{(L)})$, which sets the natural time/step-size scaling in kernel gradient descent (one can compensate by rescaling the learning rate in the idealized NTK dynamics).*

Remark 19 (Can the smallest eigenvalue become negative?) *No: for any finite network, the empirical NTK Gram matrix $\hat{K}$ at initialization is a Gram matrix of gradients, $\hat{K}_{ij} = \langle \nabla_\theta f(x_i), \nabla_\theta f(x_j) \rangle$, hence $\hat{K} \succeq 0$ deterministically and all its eigenvalues are ≥ 0 . Moreover, centering preserves positive semidefiniteness: $H\hat{K}H \succeq 0$ since H is an orthogonal projection.*

What can happen is that the deterministic proxy lower bound (e.g. $\lambda_{\min}(HK^{(L)}H) \asymp 1/(rL)$ in Proposition 16) is smaller than the magnitude of random fluctuations. In general, Weyl’s inequality gives

$$\lambda_{\min}(H\hat{K}H) \geq \lambda_{\min}(HKH) - \|H(\hat{K} - K)H\|,$$

so if $\|H(\hat{K} - K)H\|$ is larger than $1/(rL)$, this bound becomes vacuous (it may be negative), but the true eigenvalue still satisfies $\lambda_{\min}(H\hat{K}H) \geq 0$. Thus, the relevant concern is not “going below zero” but rather “being driven close to zero” when the $1/(rL)$ signal is swamped by noise.

Remark 20 (General datasets: heuristic vs theorem) Proposition 16 is a theorem under an explicit (equicorrelated) model. For general datasets with heterogeneous $\rho_{1,ij}$ and non-constant norms, translating the diagonal–off-diagonal gap of the scalar recursion into a uniform lower bound on $\lambda_{\min}(H K H)$ requires additional explicit assumptions (e.g. norm regularity and coherence bounds on $\rho_{1,ij}$). Absent such assumptions, statements of the form “ $\lambda_{\min}(H K H) \asymp 1/(rL)$ ” should be treated as heuristic rather than as part of the main theorem chain.

Remark 21 (Infinite-depth limit: existence and degeneracy after centering) The correlation recursion satisfies $1 - \rho_k = O(k^{-2})$ as in full-width MLPs at the EOC, so deep features become increasingly aligned. For the mean RF-LR NTK recursion (11), Theorem 15 implies that for any fixed $\rho_1 \in (-1, 1)$,

$$\Theta^{(L)}(\rho_1) \rightarrow \Theta_\star(r) \quad \text{and} \quad \Theta^{(L)}(1) \rightarrow \Theta_\star(r),$$

so the entrywise infinite-depth limit of the (mean) Gram matrix is a constant kernel: $K^{(L)} \rightarrow \Theta_\star(r) \mathbf{1}\mathbf{1}^\top$. Consequently, after centering one has $H K^{(L)} H \rightarrow 0$, and the relevant spectrum on $\mathbf{1}^\perp$ collapses. In particular, in the equicorrelated model Proposition 16 gives $\lambda_{\min}(H K^{(L)} H) = \Theta(1/(rL))$, so a non-degenerate deep limit on $\mathbf{1}^\perp$ can only hold under a joint scaling (e.g. r growing with L) or after an explicit rescaling, such as considering $rL H K^{(L)} H$, whose nonzero eigenvalues are $rL \lambda_\perp(K^{(L)}) \in [c_\Delta, C_\Delta]$ for all large L .

5. Low-rank preserves RKHS

In the NTK regime, training dynamics linearize: $f_t(x) \approx f_0(x) + \langle \nabla_\theta f_0(x), \theta_t - \theta_0 \rangle$. The learned function stays in the RKHS induced by Θ , with norm:

$$\|f\|_{\mathcal{H}_\Theta}^2 = \langle f, \Theta^{-1} f \rangle_{L^2}.$$

For RF-LR with frozen random features, the RKHS is characterized as follows. Universal approximation holds due to full support of $w^{(\ell)}$ (frozen Gaussian draws), so any function in the Barron space E et al. (2021) can be approximated with error $O(1/\sqrt{N})$. So the RKHS is the same as for MLPs NTK analysis shows weights are nearly frozen through training. We tackle directly this remark by freezing the weights instead of performing weights-deviation analysis on MLPs. This also removes the EOC condition on the bias.

We can now state a result highly similar to Bietti and Bach (2021): even for in the extensive rank regime, RF-LR NTK under the NTK perspective, deeper architectures do not yield a larger RKHS than shallow networks.

Mean RKHS and random features. Each realization ω of the random features yields a kernel $K_\omega(x, x') = \Theta_\omega^{(2)}(x, x')$ and an induced RKHS $\mathcal{H}_\omega$. The mean kernel $\tilde{K}(x, x') = \mathbb{E}_\omega[K_\omega(x, x')]$ is deterministic; its RKHS $\mathcal{H}_{\tilde{K}}$ is the mean RKHS. It is not $\mathbb{E}[\mathcal{H}_\omega]$ (expectation of Hilbert spaces is undefined); rather, $\mathcal{H}_{\tilde{K}}$ is the RKHS of $\mathbb{E}[K_\omega]$. For each ω , the features are $\varphi_\omega(x) = K_\omega(\cdot, x) \in \mathcal{H}_\omega$; these are random. The mean kernel has a canonical feature map $\varphi_{\tilde{K}}(x) = \tilde{K}(\cdot, x) \in \mathcal{H}_{\tilde{K}}$, which is deterministic. We have $\tilde{K}(x, x') = \mathbb{E}_\omega[\langle \varphi_\omega(x), \varphi_\omega(x') \rangle_{\mathcal{H}_\omega}]$, but $\varphi_{\tilde{K}}(x) \neq \mathbb{E}[\varphi_\omega(x)]$ in general (they live in different spaces). The mean RKHS describes the “typical” function space when averaging over random feature realizations; our result that $\mathcal{H}_{\tilde{K}}$ coincides with the shallow ReLU RKHS implies that depth does not expand this typical expressiveness.

5.1. Rank-driven concentration in RKHS (Gaussian norms product)

Having established that low rank and depth confers no RKHS advantage, we now turn to quantifying how the low-rank structure controls kernel fluctuations. We can show that the radial component of the RF-LR NTK concentrates exponentially fast in the rank r , yielding high-dimensional control of RKHS fluctuations. To state this result, we first require a homogeneity assumption on the activation function.

Assumption 1 (Homogeneous activation) *The activation function σ is positively homogeneous: $\sigma(\alpha z) = \alpha \sigma(z)$ for all $\alpha \geq 0$ and $z \in \mathbb{R}$. In particular, ReLU satisfies this property.*

We can now state our main concentration result.

Theorem 22 (Concentration in rank for homogeneous activation) *Under homogeneous activation assumption (Assumption 2.1), let $x_1, y_1 \in \mathbb{R}^r$ be the rank- r Gaussian projections of inputs x, y , with arbitrary fixed correlation $\rho \in [-1, 1]$. Define*

$$W = \frac{\|x_1\| \|y_1\|}{r}.$$

Then for any $\epsilon \in (0, 1)$,

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq 4 \exp\left(-\frac{r \epsilon^2}{8}\right).$$

Moreover, for larger deviations $\epsilon \geq 1$, there exist absolute constants $c_1, c_2 > 0$ such that

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq c_1 \exp(-c_2 r \epsilon).$$

In particular, the radial component of the RF-LR NTK concentrates exponentially fast in r , yielding high-dimensional control of RKHS fluctuations.

We now provide an intuitive explanation of the proof strategy.

Proof intuition. Set $U = \|x_1\|/\sqrt{r}$ and $V = \|y_1\|/\sqrt{r}$. U and V concentrate around 1 with sub-Gaussian tails $\exp(-r\delta^2)$. Using $|UV - 1| \leq |U - 1| + |V - 1| + |U - 1||V - 1|$ and a union bound with $\delta = \epsilon/2$ yields the stated $4 \exp(-r\epsilon^2/8)$ bound. The full proof is given in Appendix B.2.

Remark 23 *The result avoids direct manipulation of Kibble’s bivariate chi-square density, relying instead on Gaussian isoperimetry for norms and a simple algebraic bound for products.*

5.2. three-layer NTK concentration

We can now apply this concentration result to obtain bounds for the three-layer NTK.

Corollary 24 (Concentration bound for three-layer NTK) *Under Assumption 2.1, let $\hat{\Theta}_r^{(2)}(x, x') = \Psi(\hat{\rho}_r, \hat{w}_r)$ be the empirical three-layer NTK, where $\hat{\rho}_r$ is the sample correlation and $\hat{w}_r = \|x_1\| \|y_1\|/r$, with $x_1, y_1 \in \mathbb{R}^r$ being the rank- r Gaussian projections of inputs x, y with population correlation $\rho \in [-1, 1]$. Let $K_\infty(\rho) = \Theta^{(1)}(\rho)(1 - \arccos(\rho)/\pi) + \Sigma^{(1)}(\rho) + 1$ be the deterministic kernel limit.*

Then for any $\epsilon \in (0, 1)$, there exist constants $C_1, C_2 > 0$ depending on ρ and the kernel smoothness such that

$$\mathbb{P}\left(\left|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)\right| \geq \epsilon\right) \leq C_1 \exp(-C_2 r \epsilon^2).$$

Proof: See Appendix B.3.

Remark 25 (Critical cancellation of $1/r$ corrections) While the general concentration rate is $O_p(1/\sqrt{r})$ via Fisher’s CLT [Fisher \(1915\)](#); [Hotelling \(1953\)](#), the randomness induced from the gaussian process low rank output removes the $1/r$ correction term in $\Sigma^{(2)}$ that would normally appear through the radial component $\hat{w}_r$. This cancellation is critical: it is precisely the path where finite-width corrections $1/N$ analysis [Dyer and Gur-Ari \(2019\)](#) and Feynman diagram decay come into play, proving that the fluctuation rate is $1/r$ rather than $1/\sqrt{r}$. The $1/r$ scaling in the NTK recursion (Section 2) stems from the EOC parameterization and finite bottleneck width; see the “Origin of $1/r$ ” paragraph there.

Remark 26 (Extension to general L layers) The extension of this concentration bound to general depth L layers under Assumption 2.1 requires controlling nested conditional means through L layers and managing a quadratic number $O(L^2)$ of cross-terms arising from the recursive NTK structure. The technical challenge lies in tracking how the random correlations ρ_ℓ and norm products $\|h^{(\ell)}(x)\| \|h^{(\ell)}(x')\|$ propagate through layers while maintaining exponential concentration rates. This analysis is deferred to a future version.

5.3. Mean NTK over Fisher and Kibble distributions

Having established concentration results, we now turn to the mean NTK. The key observation is that the empirical NTK depends on random variables that follow Fisher [Fisher \(1915\)](#); [Hotelling \(1953\)](#) and Kibble distributions. The mean NTK is obtained by taking expectations over these distributions.

For the three-layer case, the empirical NTK is:

$$\hat{\Theta}_r^{(2)}(x, x') = 1 + \frac{1}{r} \Theta^{(1)}(\rho) \cdot \left(1 - \frac{\arccos(\hat{\rho}_r)}{\pi}\right) + \frac{1}{r} \cdot \Sigma^{(1)}(\hat{\rho}_r) \cdot \|x_1\| \|y_1\|,$$

where $\hat{\rho}_r$ follows Fisher’s distribution [Fisher \(1915\)](#); [Hotelling \(1953\)](#) (centered at ρ) and $\|x_1\| \|y_1\|$ follows Kibble’s distribution. The mean NTK is therefore:

$$\tilde{\Theta}^{(2)}(\rho) = \mathbb{E}_{\text{Fisher, Kibble}} \left[\hat{\Theta}_r^{(2)}(x, x') \right] = 1 + \frac{1}{r} \Theta^{(1)}(\rho) \cdot \mathbb{E} \left[1 - \frac{\arccos(\hat{\rho}_r)}{\pi} \right] + \frac{1}{r} \cdot \mathbb{E} \left[\Sigma^{(1)}(\hat{\rho}_r) \cdot \|x_1\| \|y_1\| \right],$$

where the expectations are taken over the Fisher distribution of $\hat{\rho}_r$ and the Kibble distribution of $(\|x_1\|^2, \|y_1\|^2)$.

Corollary 27 (Mean NTK over Fisher–Kibble has same RKHS: three-layer case) Under Assumption 2.1 and the RF-LR setting with ReLU nonlinearity and isotropic random features on $\mathbb{S}^{d-1}$, the mean NTK $\tilde{\Theta}^{(2)}$ for the three-layer case (obtained by taking expectations over Fisher and Kibble distributions) is a zonal kernel on $\mathbb{S}^{d-1}$ that induces the same RKHS as the shallow ReLU kernel.

In particular, the mean NTK RKHS coincides with the unconditional RKHS as sets with equivalent norms.

The mean NTK is:

$$\tilde{\Theta}^{(2)}(\rho) = 1 + \frac{1}{r} \Theta^{(1)}(\rho) \cdot \mathbb{E}_{\hat{\rho}_r \sim \text{Fisher}(\rho, r)} \left[1 - \frac{\arccos(\hat{\rho}_r)}{\pi} \right] + \frac{1}{r} \cdot \mathbb{E}_{\hat{\rho}_r, \|x_1\|, \|y_1\|} \left[\Sigma^{(1)}(\hat{\rho}_r) \cdot \|x_1\| \|y_1\| \right],$$

where the expectations are over Fisher [Fisher \(1915\)](#); [Hotelling \(1953\)](#) and Kibble distributions. This yields a zonal kernel with a Puiseux expansion that differs from the deterministic kernel $K_\infty(\rho)$, but preserves the same $t^{1/2}$ endpoint behavior, ensuring RKHS equivalence. The extension to general depth $L \geq 3$ is conjectured; see [Appendix B.4](#).

Hypergeometric proof of the Puiseux expansion. The corollary is non-trivial because it requires computing $\mathbb{E}[\arccos(\hat{\rho}_r)]$ for $\hat{\rho}_r \sim \text{Fisher}(\rho, r)$, integrating over Fisher’s density near $\rho = 1$. Set $\rho = 1 - t$ and $v = 1 - u$. The Fisher density is

$$p_{\text{Fisher}}(u \mid \rho, r) = \frac{(r-2) \Gamma(r-1) (1-\rho^2)^{\frac{r-1}{2}} (1-u^2)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1-\rho u)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho u}{2}\right).$$

Near $t, v \rightarrow 0^+$: $1 - \rho^2 \sim 2t$, $1 - u^2 \sim 2v$, $1 - \rho u \sim t + v$. For ${}_2F_1(\frac{1}{2}, \frac{1}{2}; r - \frac{1}{2}; z)$ with $z = 1 - \frac{t+v}{2} \rightarrow 1^-$, the connection formula (DLMF §15.8.2 [Olver et al. \(2024\)](#)) gives ${}_2F_1 = C_1(r) + O((t+v)^{r-3/2})$ since $c-a-b = r-\frac{3}{2} > 0$ (no log). With $\arccos(1-v) = \sqrt{2v} + O(v^{3/2})$ and the substitution $w = v/t$, one obtains

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} I(r) + O(t^{3/2}), \quad I(r) = \frac{(r-2) 2^{r-\frac{5}{2}}}{\sqrt{2\pi}} \frac{\Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2} - 1)}{\Gamma(r-1)} C_1(r), \quad C_1(r) = \frac{\Gamma(r-\frac{1}{2}) \Gamma(r-\frac{3}{2})}{\Gamma(r-1)^2}.$$

This yields the Puiseux expansion (5.4). By Theorem 1 of [Bietti and Bach \(2021\)](#), the $t^{1/2}$ leading exponent determines RKHS spectral decay; since it is preserved, the mean NTK induces the same RKHS as the shallow ReLU kernel.

Proposition 28 ($I(r) \sim 1/\sqrt{r}$) $I(r) \sim C/\sqrt{r}$ as $r \rightarrow \infty$. By Stirling and Beta asymptotics, $I(r) = O(1/\sqrt{r})$.

Link to concentration. Since $I(r) \sim 1/\sqrt{r}$, the Fisher–Kibble correction in the $t^{1/2}$ coefficient is $O(1/(r\sqrt{r}))$: the mean NTK approaches the deterministic limit faster as r grows, tightening the concentration bound in [Corollary 24](#).

5.4. Spectral decay and Puiseux expansions

We apply the spectral decay theory of [Bietti and Bach \(2021\)](#) to characterize the RKHS via endpoint expansions.

Near $\rho = 1$ (for $t \geq 0$), the mean NTK admits the Puiseux expansion

$$\tilde{\Theta}^{(2)}(1-t) = 1 + \frac{1}{r} (K_\infty(1) - 1) - \frac{1}{r} \left[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r) \right] t^{1/2} + O(t^{3/2}).$$

where the leading non-polynomial term has exponent $\nu = 1/2$ with coefficient $c_1(r) = -\frac{1}{r} [2/\pi + (\sqrt{2}/(2\pi)) I(r)]$; the $1/r$ prefactor comes from the NTK recursion (Section 2). Similarly, near $\rho = -1$, by rotational symmetry and homogeneity of ReLU, the expansion has the same structure with $c_{-1}(r) = c_1(r)$ (see [Bietti and Bach \(2021\)](#)).

We can now apply Theorem 1 of [Bietti and Bach \(2021\)](#) to obtain the spectral decay characterization:

Theorem 29 (Spectral decay for mean RF-LR NTK with final constant) *Let $\kappa(\rho) = \tilde{\Theta}^{(2)}(\rho)$ be the mean NTK on $\mathbb{S}^{d-1}$. From the Puiseux expansion in Appendix (see the explicit constant at the end of Appendix B.4), κ admits, for $t \geq 0$,*

$$\kappa(1-t) = p_1(t) + c_1(r) t^{1/2} + o(t^{1/2}), \quad \kappa(-1+t) = p_{-1}(t) + c_{-1}(r) t^{1/2} + o(t^{1/2}),$$

with exponent $\nu = 1/2$. For ReLU and normalized inputs,

$$c_1(r) = -\frac{1}{r} \left[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r) \right], \quad I(r) = \frac{(r-2) 2^{r-\frac{5}{2}} \Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2}-1) \cdot \Gamma(r-\frac{1}{2}) \Gamma(r-\frac{3}{2})}{\sqrt{2\pi} \Gamma(r-1) \Gamma(r-1)^2},$$

and, by symmetry of the ReLU kernel on $\mathbb{S}^{d-1}$, $c_{-1}(r) = c_1(r)$.

Then by Theorem 1 of [Bietti and Bach \(2021\)](#), the eigenvalues μ_k (in the spherical harmonics basis) satisfy:

$$\mu_k \sim C(d, \nu) k^{-d-2\nu+1} = C(d) k^{-d-2(1/2)+1} = C(d) k^{-d},$$

where $C(d) > 0$ depends only on d . Since $c_1(r) = c_{-1}(r)$, we have $|c_1(r)| = |c_{-1}(r)|$, hence:

- For even k : $\mu_k \sim (c_1(r) + c_{-1}(r)) C(d) k^{-d} = 2 c_1(r) C(d) k^{-d}$.
- For odd k : since $c_1(r) - c_{-1}(r) = 0$, $\mu_k = o(k^{-d})$ (faster-than-polynomial in this parity).

Therefore, the dominant eigenvalue decay rate is k^{-d} , determined by the even-parity eigenvalues.

Corollary 30 (No advantage in RKHS: three-layer case) *Under the RF-LR setting with ReLU nonlinearity, isotropic random features on the sphere $\mathbb{S}^{d-1}$, and in the kernel (NTK) regime, the three-layer mean NTK $\tilde{\Theta}^{(2)}$ induces the same RKHS as the shallow ReLU kernel. In particular, the corresponding RKHSs coincide as sets with equivalent norms. This is a direct application of Theorem 1 in [Bietti and Bach \(2021\)](#). The extension to depth $L \geq 3$ is conjectured (Appendix B.4).*

Remark 31 *This theorem and corollary are direct applications of [Bietti and Bach \(2021\)](#) to the RF-LR setting. Theorem 29 supplies explicit rates for approximation and generalization in the RF-LR NTK RKHS and, together with the explicit kernel formulas above, justifies that the three-layer mean NTK does not enlarge the RKHS beyond the shallow ReLU case (Corollary 30).*

6. RF-LR Gram Marchenko–Pastur spectrum for three-layer case

We now study the spectrum of the RF-LR NTK Gram matrix at initialization in a random-matrix scaling regime. This section concerns the *uncentered* Gram matrix and its spike–bulk limit; we also explain how centering removes the rank-one spike and isolates the spectrum on $\mathbf{1}^\perp$, which is the relevant subspace after centering targets (or for mean-zero targets). We work under strong scaling and data assumptions for tractability.

6.1. General setup and scaling

Assumption 2 (Random-matrix scaling) *As the number of data points $n \rightarrow \infty$:*

- *Hidden widths scale as $N \sim n^2$ (infinite-width linearization regime).*
- *Rank scales as $r \asymp n$ with ratio $\gamma_{\text{ratio}} \equiv \lim_{n \rightarrow \infty} \frac{n}{r} \in (0, \infty)$ (Marchenko–Pastur regime).*
- *Input dimension scales linearly with rank: $d \asymp r$ (so $d \asymp n$ as well).*

We now introduce the random-matrix theory conditions that are necessary for the asymptotic analysis. Those strong assumptions can be removed in future work using techniques from [Benigni and Paquette \(2025\)](#)

Assumption 3 (RMT Gaussian data) *Following the framework of [El Karoui \(2010\)](#), we impose the following conditions as $d \rightarrow \infty$:*

- (d) **Data generation:** *Input vectors satisfy $X_i = \Sigma_d^{1/2} Y_i$, where $\Sigma_d^{1/2}$ is the positive semi-definite square root of Σ_d .*
- (e) **Entrywise conditions:** *The entries of Y_i , a d -dimensional random vector, are i.i.d. Denoting by $Y_i^{(k)}$ the k -th entry of Y_i , we assume $\mathbb{E}[Y_i^{(k)}] = 0$, $\text{var}(Y_i^{(k)}) = 1$, and $\mathbb{E}[|Y_i^{(k)}|^{4+\varepsilon}] < \infty$ for some $\varepsilon > 0$ (i.e., Y_i has $4 + \varepsilon$ absolute moments).*
- (b) **Covariance structure:** *Let Σ_d be the $d \times d$ covariance matrix of the input distribution. Then Σ_d is positive semi-definite and its operator norm $\|\Sigma_d\|_2 = \sigma_1(\Sigma_d)$ remains bounded in d , that is, there exists $K > 0$ such that $\sigma_1(\Sigma_d) \leq K$ for all d .*
- (c) **Trace normalization:** *Σ_d/d has a finite limit, that is, there exists $\ell \in \mathbb{R}$ such that $\lim_{d \rightarrow \infty} \text{trace}(\Sigma_d)/d = \ell$.*

With these scaling assumptions in place, we can now analyze the microscopic structure of the kernel. The key insight is that the randomness in the low-rank projections decomposes into independent angular and radial components.

To apply [El Karoui \(2010\)](#), the kernel function $K_\infty(\rho) = \Theta^{(1)}(\rho)(1 - \arccos(\rho)/\pi) + \Sigma^{(1)}(\rho) + 1$ must be C^1 in a neighborhood of $\ell = \lim_{d \rightarrow \infty} \text{trace}(\Sigma_d)/d$ and C^3 in a neighborhood of 0. This holds trivially since $\arccos(\rho)$ is C^∞ on $(-1, 1)$ and the ReLU kernels $\Theta^{(1)}(\rho)$ and $\Sigma^{(1)}(\rho)$ have the required smoothness properties.

6.2. Signal–noise decomposition of the empirical NTK

Theorem 32 (Asymptotic normality of the low-rank kernel) *Let $x, y \in \mathbb{R}^r$ be rank- r Gaussian projections with population correlation ρ . Define*

$$\Psi(u, w) = \Theta^{(1)}(u) \left(1 - \frac{\arccos(u)}{\pi} \right) + w \Sigma^{(1)}(u) + 1,$$

so that $\hat{\Theta}_r^{(2)} = 1 + \frac{1}{r}(\Psi(\hat{\rho}_r, \hat{w}_r) - 1)$ from the $1/r$ recursion, where $\hat{\rho}_r$ is the sample correlation and $\hat{w}_r = \|x\| \|y\|/r$. Define the deterministic limit $K_\infty(\rho) = \Psi(\rho, 1)$. Under Assumption 2, as $r \rightarrow \infty$,

$$\hat{\Theta}_r^{(2)} = \Theta_{\text{obs}}^{(2)}(\rho) \xrightarrow{d} K_\infty(\rho) + \frac{1}{\sqrt{r}} \mathcal{G}(\rho) + o(r^{-1/2}),$$

where $K_\infty(\rho) = \mathbb{E}[\Theta_{\text{obs}}^{(2)}(\rho)]$ is the deterministic limit and $\mathcal{G}(\rho) \sim \mathcal{N}(0, \sigma_{\text{tot}}^2(\rho))$ is a centered Gaussian field with variance

$$\sigma_{\text{tot}}^2(\rho) = \text{Var}(\mathcal{G}(\rho)) = \underbrace{[\partial_\rho K_\infty(\rho) (1 - \rho^2)]^2}_{\text{Fisher (angle) Fisher (1915); Hotelling (1953)}} + \underbrace{[\Sigma^{(1)}(\rho) \sqrt{1 + \rho^2}]^2}_{\text{Kibble (norms)}}.$$

Proof intuition. By Fisher–Kibble decoupling (Lemma 8) Fisher (1915); Hotelling (1953), the angular $\hat{\rho}_r$ and radial $\hat{w}_r$ components are independent. A joint CLT gives $\sqrt{r}(\hat{\rho}_r - \rho, \hat{w}_r - 1) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \text{diag}((1 - \rho^2)^2, 1 + \rho^2))$. Applying the Delta method to Ψ at $(\rho, 1)$ with gradient $(\partial_\rho K_\infty(\rho), \Sigma^{(1)}(\rho))$ yields the stated variance (no cross term by independence). The limit is a deterministic kernel plus $O_p(r^{-1/2})$ Gaussian fluctuation.

Proof: See Appendix C.1.

Having established the entrywise asymptotic distribution, we now turn to the macroscopic spectral properties of the kernel matrix. The eigenvalue distribution follows a deformed Marchenko–Pastur law with a characteristic spike-bulk structure.

6.3. Macroscopic spectrum: deformed Marchenko–Pastur law

Recent work by Benigni and Paquette (2025) establishes the eigenvalue distribution of the NTK in the quadratic scaling regime ($N \sim n^2$) for single-layer networks, showing convergence to a deformed Marchenko–Pastur law with a spike-bulk structure. We extend this analysis to the three-layer case with random features and rank- r bottlenecks, where the frozen feature weights and low-rank structure modify both the spike location and bulk support.

Centered vs uncentered Gram matrices (removing the rank-one spike). Let $K \in \mathbb{R}^{n \times n}$ be a symmetric Gram matrix built from a kernel evaluated on a dataset $\{x_i\}_{i=1}^n$. Define the centering matrix $H = I_n - \frac{1}{n} \mathbf{1} \mathbf{1}^\top$ and the centered Gram matrix $K_c = H K H$. Then for any $v \perp \mathbf{1}$ we have $Hv = v$ and therefore

$$v^\top K_c v = v^\top H K H v = v^\top K v.$$

Consequently, the eigenvalues of K restricted to the subspace $\mathbf{1}^\perp$ coincide with the (nonzero) eigenvalues of K_c . In particular, if K contains a rank-one component aligned with $\mathbf{1}$ (e.g. arising from an additive constant term in the kernel), this produces a large top eigenvalue in the uncentered spectrum but is removed by centering. For kernel regression with mean-zero targets (or after centering labels), convergence rates are governed by the spectrum on $\mathbf{1}^\perp$.

Theorem 33 (Spike + bulk spectral limit) Let $\mathbf{M} = [\Theta_{ij}^{(2)}]_{i,j=1}^n$. Under Assumptions 2 and 3, as $n \rightarrow \infty$, the empirical spectrum of $\mathbf{M}$ converges to a deformed Marchenko–Pastur law with:

- A rank-one outlier (spike) of order $O(n)$: $\lambda_{\text{spike}} \approx n \alpha$, with $\alpha = K_\infty(0)$.
- A continuous bulk of order $O(1)$ supported on

$$\left[\gamma + \beta (1 - \sqrt{\gamma_{\text{ratio}}})^2, \gamma + \beta (1 + \sqrt{\gamma_{\text{ratio}}})^2 \right],$$

where $\beta = K'_\infty(0)$ and $\gamma = K_\infty(\ell) - \alpha - \ell \beta$ (diagonal shift; ℓ denotes the kernel’s diagonal argument).

Proof: See Appendix C.2 (via El Karoui (2010)). This theorem is stated for the deterministic inner-product kernel matrix built from the limiting kernel function K_∞ ; incorporating finite- r Fisher–Kibble fluctuations or finite-width effects into the limiting ESD would require BP-style resolvent/Gaussian-equivalence machinery and is left for future work.

Remark 34 (How to interpret the spike for learning) The spike in Theorem 33 is a statement about the uncentered Gram matrix. When the kernel includes an additive constant term (here the $+1$ in K_∞), the associated rank-one component aligns with $\mathbf{1}$ and yields $\lambda_{\max}(\mathbf{M}) = \Theta(n)$. This does not by itself imply poor conditioning of the learning problem: after centering (or for mean-zero labels), the relevant spectrum is that of HMMH on $\mathbf{1}^\perp$, i.e. the bulk/shifted spectrum.

Remark 35 (Bulk laws vs extreme eigenvalues) Theorem 33 is a statement about the empirical spectral distribution (a bulk law) under strong random-matrix assumptions. Even when a limiting bulk measure exists and has compact support, this does not automatically control the extreme eigenvalues at finite n : the smallest eigenvalue can be sensitive to (i) finite-size edge fluctuations; (ii) deviations from the independence/isotropy assumptions; and (iii) additional structured components beyond the rank-one spike (e.g. heteroskedastic norms, anisotropic covariance Σ_d , or higher-order kernel terms).

Centering removes only the rank-one component aligned with $\mathbf{1}$ and leaves the spectrum on $\mathbf{1}^\perp$ unchanged (paragraph “Centered vs uncentered”); it does not make the remaining eigenvalue distribution “less skewed”. In particular, in depth-scaling regimes where the relevant centered scale becomes small (e.g. $\lambda_{\min}(HK^{(L)}H) \asymp 1/(rL)$ in the equicorrelated proxy of Section 4), random fluctuations can drive $\lambda_{\min}(H\hat{K}H)$ very close to 0 even though $\hat{K} \succeq 0$ always holds (see Remark 19).

Controlling the smallest eigenvalue beyond bulk limits typically requires separate “local law” or edge universality inputs, which are outside the scope of the present random-matrix section.

We can summarize the three-layer results as follows.

three-layer case summary. Specializing to two layers with ReLU features, the deterministic kernel limit is $K_\infty(\rho) = \Theta^{(1)}(\rho)(1 - \arccos(\rho)/\pi) + \Sigma^{(1)}(\rho) + 1$. At the dataset level, the corresponding Gram matrix exhibits a rank-one “spike” of size $\lambda_{\text{spike}} \approx n K_\infty(0)$, superimposed on a deformed Marchenko–Pastur bulk. The bulk support is the interval given by Theorem 33, with slope $\beta = K'_\infty(0)$ and diagonal shift γ determined by the kernel’s on-diagonal value.

7. Conclusion

This work develops an NTK analysis for low-rank random-feature architectures (RF-LR/MMNN) under explicit assumptions. **Expressivity (RKHS).** In the NTK regime, the three-layer RF-LR mean kernel induces the same RKHS as the shallow ReLU kernel (Corollary 30), so depth does not enlarge the RKHS in this setting. **Concentration in the bottleneck dimension.** For the two-layer low-rank step, the empirical kernel concentrates around its deterministic limit $K_\infty(\rho)$ with tails controlled by the bottleneck dimension r (Theorem 22, Corollary 24). **Spectrum (three-layer case).** Under random-matrix scaling assumptions, the Gram matrix exhibits a spike–bulk structure and a deformed Marchenko–Pastur limit (Theorem 33), while the entrywise fluctuations satisfy a CLT with separate angular (Fisher) and radial (Kibble) contributions (Theorem 32). **Recursion.**

We provide an explicit NTK recursion for arbitrary depth in RF-LR and a closed-form expansion (Theorem 3, Corollary 4), which can be used as a starting point for finite-width corrections.

8. Discussion

This work leaves several natural directions open. We emphasize that our current scope is the lazy/NTK regime under explicit simplifying assumptions; extending beyond this setting requires additional ideas.

A technical feature of our setup is that freezing the random features removes some cross-layer dependencies that appear in fully trained networks. Combined with Fisher–Kibble decoupling Fisher (1915); Hotelling (1953), this makes certain calculations more explicit and suggests a path toward relaxing some of our random-matrix assumptions using techniques in Benigni and Paquette (2025).

The recursion (Theorem 3) suggests organizing finite-width corrections as an expansion in the bottleneck dimension r . Making this program rigorous for general depth would require controlling the growth of cross-terms produced by the recursion and propagating concentration through the composed Gaussian-process layers.

On the theoretical front, several important questions remain open and constitute an immediate research agenda. The extension of the exponential concentration bounds to general depth L layers requires controlling nested conditional means through L layers and managing a quadratic number $O(L^2)$ of cross-terms arising from the recursive NTK structure (Remark after Corollary 24). The technical challenge lies in tracking how the random correlations ρ_ℓ and norm products $\|h^{(\ell)}(x)\| \|h^{(\ell)}(x')\|$ propagate through layers while maintaining exponential concentration rates. A **non-Gaussian process propagation analysis** distinct from NNGP fits would provide complementary insights beyond the kernel regime, potentially revealing how feature learning emerges from deviations from Gaussianity. An **explicit bulk formula** with normalization scaled by r^2 would complete the spectral picture and enable direct comparisons with Marchenko–Pastur predictions. Relaxing the strong assumptions in Assumptions 2 and 3 (Gaussian data, specific scaling relationships) using techniques from Benigni and Paquette (2025) would extend the applicability of our spectral results to more realistic data distributions.

Additional theoretical extensions hold particular promise. The current analysis focuses on the three-layer case for tractability; extending to arbitrary depth L while maintaining explicit formulas remains an open challenge. Investigating how the edge-of-chaos (EOC) parameterization affects finite-width corrections and the Nr scaling mentioned in the architecture definition would provide concrete quantitative bounds. Understanding how the curse of dimensionality impacts normalization choices (unit norm weights vs. biases) depending on data geometry (hypercube vs. spherical) would provide principled guidance for practitioners.

The experimental validation of our theoretical predictions constitutes an equally important and concrete next step. We envision a comprehensive program that includes **confirming lazy-kernel predictions** and precisely mapping the boundary of lazy training in practice; conducting systematic **three-layer width sweeps** to test the $O(1/n) + O(1/r)$ finite-width corrections suggested by the expansion and identify minimal (n, r) configurations where kernel predictions become accurate; and a detailed **comparison suite with MLPs** that probes NTK eigenvalue distributions, validates the spike-bulk structure, and measures task accuracy at matched parameter counts. These studies would verify the **exponential concentration** rates (Theorem 22), validate the **single-outlier struc-**

ture (Theorem 33), and demonstrate the **practical accessibility of the kernel regime** for low-rank random features.

Finally, it would be valuable to connect the low-rank random-feature recursion studied here with broader frameworks for infinite-width limits (e.g. tensor programs Yang (2020)) and with depth-dependent NTK spectral analyses for MLPs at the edge of chaos Terjék and González-Sánchez (2025), while maintaining explicit control of the bottleneck scaling.

Our present scope is deliberately focused on the **lazy/NTK regime**, where analytical tractability is greatest and the advantages of low-rank structure are most clearly visible. We believe that completing the pending finite-width formulas, spectrum characterizations, and experimental validations will turn this into a comprehensive NTK account of RF-LR networks and establish a solid foundation for analyses beyond the lazy regime.

Comparison to full-width MLPs at the EOC (Terjék–González-Sánchez, 2025)

To emphasize what is specific to low-rank random-feature training, we contrast our depth/rank scalings with the full-width MLP-at-EOC results of Terjék and González-Sánchez (2025). The main structural difference is that in RF-LR only $A^{(\ell)}$ is trained and the NTK recursion carries an explicit $1/r$ factor at bottlenecks (Theorem 3); in full-width MLPs, all layers are trained and the limiting EOC NTK grows linearly with depth in the uncentered spectrum.

	Full-width MLP at EOC Terjék and González-Sánchez (2025)	
Trainable parameters	all weight layers	
Key scaling parameter	depth L , activation Δ_ϕ	
Depth effect (kernel magnitude)	$\lambda_{\max}(K_{\text{EOC}}) = \Theta(L)$ (Thm. 4.1)	n
Centered spectrum	bounds on $1^\perp$ via inverse-cosine matrices	equicorrelat
Uncentered spike	rank-one component present; grows with L and n	rank-one spike al

Advantages and limitations. Relative to full-width EOC MLPs, the RF-LR recursion introduces a rank–depth tradeoff: for fixed r , depth can improve correlation alignment but the centered gap can decay as $1/(rL)$ in simple models (Prop. 16), suggesting that maintaining a uniform spectral gap with depth may require r to grow with L . Conversely, RF-LR offers a reduced trainable parameter budget ($O(rN)$) and rank-driven concentration effects, and yields explicit recursion-based formulas for depth dependence that are not available in the same form for fully trained MLPs.

Optimization interpretation in the lazy/NTK regime (stability vs speed). In kernel gradient descent on squared loss, the stability (no oscillatory divergence) constraint is controlled by the *largest* eigenvalue on the relevant invariant subspace: for the centered operator $K_c = HKH$ one needs $0 < \eta < 2/\lambda_{\max}(K_c)$, while the convergence speed is governed by $\lambda_{\min}(K_c)$ (or by the condition number on $1^\perp$). This highlights the qualitative RF-LR vs full-width difference with depth: in full-width EOC MLPs, the limiting kernel magnitude typically grows with depth L Terjék and González-Sánchez (2025), forcing the stable step size to shrink with L , whereas in RF-LR the mean recursion saturates (Theorem 15), so $\lambda_{\max}$ does not grow linearly with L in the same way. At the same time, RF-LR can exhibit a small centered scale $\lambda_{\min}(HKH) \asymp 1/(rL)$ in simple proxy models (Proposition 16), which slows convergence unless one rescales η accordingly; finite- r fluctuations can further reduce the effective gap (Remark 19).

Learning-rate/conditioning compromise (standard kernel GD bound). To make the tradeoff explicit, consider the centered kernel least-squares iteration on $\mathbf{1}^\perp$,

$$e_{t+1} = (I - \eta K_c) e_t,$$

where $K_c \succeq 0$ has eigenvalues in $[\mu, L]$ with $0 < \mu = \lambda_{\min}(K_c) \leq \lambda_{\max}(K_c) = L$. Then:

- **Stability:** $0 < \eta < 2/L$ is necessary and sufficient for $\|e_t\| \rightarrow 0$ for all initial errors e_0 .
- **Best linear rate (fixed step size):** choosing $\eta^* = 2/(L + \mu)$ yields

$$\|e_t\| \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^t \|e_0\|, \quad \kappa = \frac{L}{\mu}.$$

Thus, a smaller $L = \lambda_{\max}(K_c)$ permits a larger stable learning rate, but the achievable convergence rate still depends on the condition number κ : improving stability by controlling $\lambda_{\max}$ does not by itself remove slow directions when $\lambda_{\min}$ is small. In RF-LR depth scalings, saturation of $\lambda_{\max}$ with L can prevent learning-rate shrinkage with depth, while a decay $\mu \asymp 1/(rL)$ would imply κ growing like rL in simple proxy models (and hence iteration counts scaling accordingly) unless additional structure or preconditioning is exploited.

When a small $\lambda_{\min}$ does not hurt (pure rescaling). If changing architectural parameters (r, L) predominantly *rescales* the centered kernel, i.e. $K_c^{(r,L)} \approx s_{r,L} \tilde{K}_c$ for some scalar $s_{r,L} > 0$ and a shape matrix $\tilde{K}_c$ with comparable condition number across (r, L) , then the optimization behavior at initialization depends mainly on $\kappa(\tilde{K}_c)$, because the stable learning-rate range scales as $\eta \in (0, 2/(s_{r,L} \lambda_{\max}(\tilde{K}_c)))$. In the equicorrelated proxy model, $K_c = \lambda_\perp(L, r) H$ has $\kappa = 1$ on $\mathbf{1}^\perp$ and $\lambda_\perp(L, r) \asymp 1/(rL)$, so the dependence on the product rL is purely a *scale* effect that can be absorbed by choosing $\eta \propto rL$ (up to the stability limit). Departures from this scale-only picture (heterogeneous data geometry, non-flat spectra, and finite-width/finite- r fluctuations) are precisely where rank/depth tradeoffs become substantive.

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Appendix A. Proofs of NTK Recursions

A.1. Proof of Theorem 3: Infinite-width NTK recursion

Proof We prove the recursion by decomposing the NTK at layer ℓ into contributions of the trainable parameters up to layer ℓ , and invoking the Law of Large Numbers (LLN) under the sequential

infinite-width limit. Let $f^{(\ell)}$ denote the network output truncated at layer ℓ . For finite width, the (random) NTK reads

$$\Theta_{\text{rand}}^{(\ell)}(x, x') = \underbrace{(\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x))^\top (\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x'))}_{\text{Term 1}} + \underbrace{\sum_{j=1}^{n_\ell} \frac{\partial f^{(\ell)}(x)}{\partial A_j^{(\ell)}} \frac{\partial f^{(\ell)}(x')}{\partial A_j^{(\ell)}}}_{\text{Term 2}}. \quad (12)$$

Bias parameters. Under the NTK parameterization with trainable biases at each layer, the gradient with respect to the bias contributes an identity kernel. Concretely, for layer ℓ , the bias partial derivative satisfies $\partial f^{(\ell)}(x)/\partial b^{(\ell)} = \sigma_A/\sqrt{n_\ell}$, so the associated NTK contribution converges by LLN to a deterministic constant, which we normalize as 1 under our parameterization. This yields the additive +1 term in the recursion.

Term 2 (layer- ℓ output weights $A^{(\ell)}$). By the forward pass, $\partial f^{(\ell)}(x)/\partial A_j^{(\ell)} = \frac{\sigma_A}{\sqrt{n_\ell}} \sigma(\mathbf{w}_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x) + \beta b_j^{(\ell)})$. Therefore

$$\text{Term 2} = \frac{\sigma_A^2}{n_\ell} \sum_{j=1}^{n_\ell} \sigma(\mathbf{w}_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x) + \beta b_j^{(\ell)}) \sigma(\mathbf{w}_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x') + \beta b_j^{(\ell)}). \quad (13)$$

Conditionally on $\mathbf{h}^{(\ell-1)}$, the summands are i.i.d. over $(\mathbf{w}_j^{(\ell)}, b_j^{(\ell)})$. As $n_\ell \rightarrow \infty$ (with lower layers already taken to their limits), LLN yields

$$\text{Term 2} \xrightarrow{p} \sigma_A^2 \Sigma^{(\ell)}(x, x'). \quad (14)$$

Term 1 (previous layers). By the chain rule,

$$\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x) = \left(\frac{\partial f^{(\ell)}(x)}{\partial \mathbf{h}^{(\ell-1)}(x)} \right) \nabla_{\theta^{(\ell-1)}} \mathbf{h}^{(\ell-1)}(x). \quad (15)$$

Hence Term 1 factorizes as $\Theta_{\text{rand}}^{(\ell-1)}(x, x')$ multiplied by a Jacobian metric:

$$\text{Term 1} = \Theta_{\text{rand}}^{(\ell-1)}(x, x') \cdot \left\langle \frac{\partial f^{(\ell)}(x)}{\partial \mathbf{h}^{(\ell-1)}(x)}, \frac{\partial f^{(\ell)}(x')}{\partial \mathbf{h}^{(\ell-1)}(x')} \right\rangle. \quad (16)$$

Taking the inner product and applying LLN as $n_\ell \rightarrow \infty$ yields

$$\left\langle \frac{\partial f^{(\ell)}(x)}{\partial \mathbf{h}^{(\ell-1)}(x)}, \frac{\partial f^{(\ell)}(x')}{\partial \mathbf{h}^{(\ell-1)}(x')} \right\rangle \xrightarrow{p} \sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x'). \quad (17)$$

Therefore,

$$\text{Term 1} \xrightarrow{p} \Theta^{(\ell-1)}(x, x') \cdot \sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x'). \quad (18)$$

Combining the bias contribution with the two limits completes the recursion. For bottleneck layers $\ell \geq 2$, the Jacobian and output contributions scale with $1/r$ (from the r -dimensional bottleneck); we absorb σ_A^2 into the normalization. Thus

$$\Theta^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'), \quad \Theta^{(\ell)}(x, x') = 1 + \frac{1}{r} \Theta^{(\ell-1)}(x, x') \cdot \sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x') + \frac{1}{r} \sigma_A^2 \Sigma^{(\ell)}(x, x') \quad (\ell \geq 2) \quad (19)$$

The sequential limit assumption ensures that lower-layer random objects have already converged when taking the next-layer limit, justifying the conditional LLN applications. $\blacksquare$

A.2. Proof of Theorem 2: Gaussian process composition

Proof We prove by induction on ℓ that $\mathbf{h}^{(\ell)}$ converges to a Gaussian process with covariance kernel $\Sigma^{(\ell)}$.

Base case $\ell = 0$. For $\ell = 0$, we have $\mathbf{h}^{(0)}(x) = x$, which is deterministic and trivially a Gaussian process.

Inductive step. Assume that $\mathbf{h}^{(\ell-1)}$ converges to a Gaussian process with covariance kernel $\Sigma^{(\ell-1)}$. Consider the forward pass at layer ℓ :

$$\mathbf{h}^{(\ell)}(x) = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \boldsymbol{\sigma} \left(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x) + b_j^{(\ell)} \right) + c. \quad (20)$$

For any finite collection of inputs $\{x_1, \dots, x_m\}$, conditionally on $\mathbf{h}^{(\ell-1)}$, the terms $A_j^{(\ell)} \boldsymbol{\sigma} \left(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x_i) + b_j^{(\ell)} \right)$ are independent across j and identically distributed. Since $A_j^{(\ell)} \sim \mathcal{N}(0, \sigma_A^2)$ and $w_j^{(\ell)}, b_j^{(\ell)}$ are frozen Gaussian draws, the conditional distribution of each term is centered with finite variance.

By the multivariate Central Limit Theorem, as $n_\ell \rightarrow \infty$, the vector $(\mathbf{h}^{(\ell)}(x_1), \dots, \mathbf{h}^{(\ell)}(x_m))$ converges in distribution to a multivariate Gaussian. The covariance is given by

$$\mathbb{E} \left[\mathbf{h}^{(\ell)}(x_i) \mathbf{h}^{(\ell)}(x_j)^\top \right] = \mathbb{E}_w \left[\boldsymbol{\sigma} \left(w^\top \mathbf{h}^{(\ell-1)}(x_i) \right) \boldsymbol{\sigma} \left(w^\top \mathbf{h}^{(\ell-1)}(x_j) \right)^\top \right] = \Sigma^{(\ell)}(x_i, x_j), \quad (21)$$

where the expectation is taken over the frozen weights w and the limiting Gaussian process $\mathbf{h}^{(\ell-1)}$. The last equality follows from the definition of the base kernel in Definition 1.

The sequential infinite-width limit ensures that $\mathbf{h}^{(\ell-1)}$ has already converged to its Gaussian process limit when taking the limit for layer ℓ , justifying the application of the CLT conditionally on $\mathbf{h}^{(\ell-1)}$. $\blacksquare$

A.3. Proof of three-layer NTK Corollary

Proof Apply Theorem 3 for $\ell = 1$ and $\ell = 2$. For $\ell = 1$, since $\Theta^{(0)} = 0$, the recursion gives:

$$\Theta^{(1)}(x, x') = 1 + \Theta^{(0)}(x, x') \cdot \dot{\Sigma}^{(1)}(x, x') + \Sigma^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'). \quad (22)$$

For $\ell = 2$, substitute $\Theta^{(1)} = 1 + \Sigma^{(1)}$ into the recursion:

$$\Theta^{(2)}(x, x') = 1 + \frac{1}{r} \Theta^{(1)}(x, x') \cdot \dot{\Sigma}^{(2)}(x, x') + \frac{1}{r} \Sigma^{(2)}(x, x'). \quad (23)$$

This is the stated three-layer recursion with the bias contribution.

Mean explicit form (low-rank). Under the homogeneous ReLU setting on the sphere, the mean three-layer NTK (with $1/r$ factors) is

$$\tilde{\Theta}^{(2)}(\rho) = 1 + \frac{1}{r} \Theta^{(1)}(\rho) \mathbb{E} \left[1 - \frac{\arccos(\hat{\rho}_r)}{\pi} \right] + \frac{1}{r} \mathbb{E} \left[\Sigma^{(1)}(\hat{\rho}_r) \|x_1\| \|y_1\| \right], \quad (24)$$

where the expectations are over Fisher and Kibble. As $r \rightarrow \infty$, $\hat{\rho}_r \rightarrow \rho$ and $\|x_1\| \|y_1\| / r \rightarrow 1$, so the limit $K_\infty(\rho) = \Theta^{(1)}(\rho)(1 - \arccos(\rho)/\pi) + \Sigma^{(1)}(\rho) + 1$ is recovered (without the $1/r$ prefactors, since the low-rank scaling is absorbed in the limit). $\blacksquare$

A.4. Proof of General L -layer NTK Corollary

Proof We prove the explicit form by induction on L , accounting for the $+1$ bias terms in the recursion.

Base case $L = 1$. From Theorem 3, $\Theta^{(1)} = 1 + \Sigma^{(1)}$. The formula with $L = 1$ gives:

$$\Theta^{(1)}(x, x') = \sum_{\ell=1}^1 \left(\prod_{k=\ell+1}^1 \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^1 \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^1 \dot{\Sigma}^{(k)}(x, x'). \quad (25)$$

For the first sum: when $\ell = 1$, $\prod_{k=2}^1 \dot{\Sigma}^{(k)} = 1$ (empty product). For the second sum: when $\ell = 1$, $\Sigma^{(1)} \cdot \prod_{k=2}^1 \dot{\Sigma}^{(k)} = \Sigma^{(1)} \cdot 1 = \Sigma^{(1)}$. Therefore, $\Theta^{(1)} = 1 + \Sigma^{(1)}$, which matches the recursion.

Inductive step. Assume the formula holds for depth $L - 1$:

$$\Theta^{(L-1)}(x, x') = \sum_{\ell=1}^{L-1} \left(\prod_{k=\ell+1}^{L-1} \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^{L-1} \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^{L-1} \dot{\Sigma}^{(k)}(x, x'). \quad (26)$$

By Theorem 3, for $L \geq 2$,

$$\Theta^{(L)}(x, x') = 1 + \frac{1}{r} \Theta^{(L-1)}(x, x') \cdot \dot{\Sigma}^{(L)}(x, x') + \frac{1}{r} \Sigma^{(L)}(x, x'). \quad (27)$$

Substituting the inductive hypothesis and distributing the $1/r$ factor yields the stated closed form; the first sum gains a new term 1 (from $\ell = L$) and each existing term is multiplied by $(1/r)\dot{\Sigma}^{(L)}$, while the second sum gains $(1/r)\Sigma^{(L)}$ and each existing term is multiplied by $(1/r)\dot{\Sigma}^{(L)}$. The first term 1 corresponds to $\ell = L$ in the first sum (with empty product $\prod_{k=L+1}^L \dot{\Sigma}^{(k)} = 1$). The last term $\Sigma^{(L)}$ corresponds to $\ell = L$ in the second sum (with empty product $\prod_{k=L+1}^L \dot{\Sigma}^{(k)} = 1$). Therefore:

$$\Theta^{(L)}(x, x') = \sum_{\ell=1}^L \left(\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^L \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x'), \quad (28)$$

completing the induction. The total number of terms is $L + L = 2L$, which grows linearly with depth. $\blacksquare$

Appendix B. Proofs of low-rank NTK RKHS

B.1. Discussion on Fisher and Kibble Distributions

Lemma 36 (Fisher–Kibble decoupling) *Let x, y be input vectors and let x_1, y_1 denote their rank- r random projections. Define the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ and squared norms $u = \|x_1\|^2, v = \|y_1\|^2$. Then:*

- ρ_1 follows Fisher’s correlation distribution [Fisher \(1915\)](#) [Hotelling \(1953\)](#), centered at the true correlation ρ .
- (u, v) follow Kibble’s [Kibble \(1941\)](#) bivariate Gamma (chi-square) law.

- *Angular and radial parts are independent:* $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) p_{\text{Kibble}}(u, v)$.

Proof Let $x, y \in \mathbb{R}^d$ be fixed input vectors with correlation $\rho = \langle x, y \rangle / (\|x\| \|y\|)$. Consider a rank- r random projection matrix $P \in \mathbb{R}^{r \times d}$ with entries $P_{ij} \sim \mathcal{N}(0, 1/d)$ i.i.d., so that $x_1 = Px$ and $y_1 = Py$ are the projected vectors.

Since P has i.i.d. Gaussian entries, the projected vectors $x_1, y_1 \in \mathbb{R}^r$ form a ρ -correlated bivariate Gaussian pair. Specifically, for each component $i = 1, \dots, r$, the pairs (x_{1i}, y_{1i}) are i.i.d. with joint distribution

$$\begin{pmatrix} x_{1i} \\ y_{1i} \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \|x\|^2/d & \rho\|x\|\|y\|/d \\ \rho\|x\|\|y\|/d & \|y\|^2/d \end{pmatrix}\right). \quad (29)$$

By rotational invariance of the Gaussian projection and the fact that the correlation ρ is preserved under orthogonal transformations, we can assume without loss of generality that the covariance structure factors into angular and radial components.

Angular part (Fisher distribution). The empirical correlation is defined as

$$\rho_1 = \frac{\langle x_1, y_1 \rangle}{\|x_1\| \|y_1\|} = \frac{\sum_{i=1}^r x_{1i} y_{1i}}{\sqrt{\sum_{i=1}^r x_{1i}^2} \sqrt{\sum_{i=1}^r y_{1i}^2}}. \quad (30)$$

This is the sample correlation coefficient of r pairs of correlated Gaussian random variables. Fisher [Fisher \(1915\)](#); [Hotelling \(1953\)](#) showed that the distribution of ρ_1 depends only on the true correlation ρ and the sample size r , with density given in Theorem 3.2. The key observation is that ρ_1 is a function purely of the angles between the projected vectors, independent of their magnitudes.

Radial part (Kibble distribution). The squared norms $u = \|x_1\|^2 = \sum_{i=1}^r x_{1i}^2$ and $v = \|y_1\|^2 = \sum_{i=1}^r y_{1i}^2$ are sums of correlated chi-square variables. Since (x_{1i}, y_{1i}) are ρ -correlated Gaussians, the bivariate distribution of (u, v) follows Kibble's [Kibble \(1941\)](#) bivariate chi-square law. The joint density involves the modified Bessel function $I_\nu(z)$, which appears because the dot product $\langle x_1, y_1 \rangle = \sum_{i=1}^r x_{1i} y_{1i}$ can be expressed as a weighted sum of products of correlated normals, whose distribution relates to Bessel functions through the generating function of the bivariate chi-square distribution.

Specifically, the joint characteristic function of (u, v) is

$$\phi(s, t) = \mathbb{E}[e^{i(su + tv)}] = (1 - 2i(1 - \rho^2)(s + t) - 4\rho^2 st)^{-r/2}, \quad (31)$$

and the inverse Fourier transform yields Kibble's density, which contains the modified Bessel function $I_{\frac{r}{2}-1}$ as shown in Lemma 3.2. The Bessel function arises from the integral representation of the bivariate chi-square density when the correlation $\rho \neq 0$.

Independence of angular and radial parts. Rotational invariance of isotropic Gaussian projections implies that $\rho_1 = \cos \angle(x_1, y_1)$ and the norms $(\|x_1\|, \|y_1\|)$ are independent, yielding the factorization $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) \cdot p_{\text{Kibble}}(u, v)$. This is the classical independence of the sample correlation from sample variances in bivariate normal data. $\blacksquare$

Lemma 37 (Nonasymptotic concentration of Gaussian sample cosine) *Let $r \geq 1$ and let $(X, Y) \in \mathbb{R}^r \times \mathbb{R}^r$ have i.i.d. coordinates $\{(X_i, Y_i)\}_{i=1}^r$, each distributed as a centered bivariate normal with $\mathbb{E}[X_i^2] = \mathbb{E}[Y_i^2] = 1$ and $\mathbb{E}[X_i Y_i] = \rho \in (-1, 1)$. Define the empirical cosine (sample correlation without centering)*

$$\hat{\rho}_r = \frac{\langle X, Y \rangle}{\|X\| \|Y\|}.$$

Then there exist absolute constants $c, C > 0$ such that for all $t \in (0, 1)$,

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq 6 \exp(-c r t^2), \quad \mathbb{E}[(\hat{\rho}_r - \rho)^2] \leq \frac{C}{r}.$$

Proof Write

$$S_{xx} = \frac{1}{r} \|X\|^2, \quad S_{yy} = \frac{1}{r} \|Y\|^2, \quad S_{xy} = \frac{1}{r} \langle X, Y \rangle, \quad \text{so that} \quad \hat{\rho}_r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}.$$

Each of $S_{xx} - 1$, $S_{yy} - 1$, and $S_{xy} - \rho$ is an average of i.i.d. centered sub-exponential random variables (since $X_i^2 - 1$, $Y_i^2 - 1$, and $X_i Y_i - \rho$ have finite ψ_1 norms for Gaussian data). Therefore, by Bernstein's inequality, there exist absolute constants $c_0 > 0$ and $C_0 < \infty$ such that for all $t \in (0, 1)$,

$$\mathbb{P}(|S_{xx} - 1| \geq t) \leq 2e^{-c_0 r t^2}, \quad \mathbb{P}(|S_{yy} - 1| \geq t) \leq 2e^{-c_0 r t^2}, \quad \mathbb{P}(|S_{xy} - \rho| \geq t) \leq 2e^{-c_0 r t^2}.$$

On the event $E_t = \{|S_{xx} - 1| \leq t, |S_{yy} - 1| \leq t\}$ with $t \leq 1/2$, we have $S_{xx}, S_{yy} \in [1/2, 3/2]$, hence

$$\left| \frac{1}{\sqrt{S_{xx} S_{yy}}} - 1 \right| \leq C_0 (|S_{xx} - 1| + |S_{yy} - 1|) \leq 2C_0 t$$

for an absolute constant C_0 . Using

$$\hat{\rho}_r - \rho = \frac{S_{xy} - \rho}{\sqrt{S_{xx} S_{yy}}} + \rho \left(\frac{1}{\sqrt{S_{xx} S_{yy}}} - 1 \right),$$

we obtain on E_t the bound $|\hat{\rho}_r - \rho| \leq C_1 (|S_{xy} - \rho| + t)$ for an absolute constant C_1 . Taking $t \asymp t'$ and applying a union bound over the three Bernstein tails yields $\mathbb{P}(|\hat{\rho}_r - \rho| \geq t') \leq 6e^{-c r t'^2}$ for $t' \in (0, 1)$ and absolute $c > 0$. The second-moment bound follows by integrating the tail bound: $\mathbb{E}[(\hat{\rho}_r - \rho)^2] = \int_0^\infty 2u \mathbb{P}(|\hat{\rho}_r - \rho| \geq u) du \leq C/r$. $\blacksquare$

B.2. Proof of Theorem 22: Rank-driven concentration

Proof Let $x_1, y_1 \in \mathbb{R}^r$ be Gaussian projections with arbitrary fixed correlation $\rho \in [-1, 1]$. Define $U = \|x_1\|/\sqrt{r}$, $V = \|y_1\|/\sqrt{r}$, and $W = UV = \|x_1\| \|y_1\|/r$. We prove that for $\epsilon \in (0, 1)$,

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq 4 \exp\left(-\frac{r\epsilon^2}{8}\right), \quad (32)$$

and that for $\epsilon \geq 1$ there exist constants $c_1, c_2 > 0$ with $\mathbb{P}(|W - 1| \geq \epsilon) \leq c_1 e^{-c_2 r \epsilon}$.

The Euclidean norm is 1-Lipschitz, so by Gaussian concentration, for all $\delta > 0$,

$$\mathbb{P}(|U - 1| \geq \delta) \leq 2 \exp\left(-\frac{r\delta^2}{2}\right), \quad \mathbb{P}(|V - 1| \geq \delta) \leq 2 \exp\left(-\frac{r\delta^2}{2}\right). \quad (33)$$

We use

$$|UV - 1| \leq |U - 1| + |V - 1| + |U - 1||V - 1|. \quad (34)$$

Fix $\epsilon \in (0, 1)$ and set $\delta = \epsilon/2$. On the event $\{|U - 1| < \delta, |V - 1| < \delta\}$, we have

$$|UV - 1| \leq 2\delta + \delta^2 \leq \epsilon. \quad (35)$$

Therefore,

$$\mathbb{P}(|UV - 1| \geq \epsilon) \leq \mathbb{P}(|U - 1| \geq \delta) + \mathbb{P}(|V - 1| \geq \delta) \leq 4 \exp\left(-\frac{r\epsilon^2}{8}\right). \quad (36)$$

■

B.3. Proof of Corollary 24: Concentration bound for three-layer NTK

Proof We prove that the empirical three-layer NTK $\hat{\Theta}_r^{(2)}(x, x') = \Psi(\hat{\rho}_r, \hat{w}_r)$ concentrates exponentially fast around its deterministic limit $K_\infty(\rho) = \Psi(\rho, 1)$ as $r \rightarrow \infty$.

Recall that the kernel function is defined as:

$$\Psi(u, w) = \Theta^{(1)}(u) \left(1 - \frac{\arccos(u)}{\pi}\right) + w \Sigma^{(1)}(u) + 1, \quad (37)$$

where $\hat{\rho}_r$ is the sample correlation and $\hat{w}_r = \|x_1\| \|y_1\| / r$. By Lemma 37, $\hat{\rho}_r$ concentrates around ρ with sub-Gaussian tails at rate $r^{-1/2}$, and by Theorem 22, $\hat{w}_r$ concentrates around 1 with sub-Gaussian tails.

The function Ψ is differentiable in both arguments. For fixed $\rho \in (-1, 1)$, we have:

$$\partial_u \Psi(\rho, 1) = \partial_\rho K_\infty(\rho) = O(1), \quad (38)$$

$$\partial_w \Psi(\rho, 1) = \Sigma^{(1)}(\rho) = O(1). \quad (39)$$

Since $\Theta^{(1)}$, $\Sigma^{(1)}$, and $\arccos$ are smooth on $(-1, 1)$, there exist Lipschitz constants $L_u, L_w > 0$ (depending on ρ) such that for $u, u' \in [\rho - \delta, \rho + \delta]$ and $w, w' \in [1 - \delta, 1 + \delta]$ with $\delta > 0$ small:

$$|\Psi(u, w) - \Psi(u', w')| \leq L_u |u - u'| + L_w |w - w'|. \quad (40)$$

We combine radial and angular concentration by using a union bound and the Lipschitz property:

$$|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| = |\Psi(\hat{\rho}_r, \hat{w}_r) - \Psi(\rho, 1)| \leq L_u |\hat{\rho}_r - \rho| + L_w |\hat{w}_r - 1|. \quad (41)$$

Therefore, for $\epsilon \in (0, 1)$:

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq \mathbb{P}\left(|\hat{\rho}_r - \rho| \geq \frac{\epsilon}{2L_u}\right) + \mathbb{P}\left(|\hat{w}_r - 1| \geq \frac{\epsilon}{2L_w}\right). \quad (42)$$

Now we prove the sub-Gaussian concentration of the sample correlation via Hanson–Wright.

Lemma 38 (Sub-Gaussian concentration of the sample correlation via Hanson–Wright) *Let $(X_i, Y_i)_{i=1}^r$ be i.i.d. centered Gaussian pairs with unit variances and correlation $\rho \in (-1, 1)$. Define*

$$S_{xx} = \frac{1}{r} \sum_{i=1}^r X_i^2, \quad S_{yy} = \frac{1}{r} \sum_{i=1}^r Y_i^2, \quad S_{xy} = \frac{1}{r} \sum_{i=1}^r X_i Y_i, \quad \hat{\rho}_r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}. \quad (43)$$

There exist absolute constants $c, C > 0$ such that for all $t \in (0, 1)$,

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq C \exp(-c r t^2). \quad (44)$$

Proof

Write $Y = \rho X + \sqrt{1 - \rho^2} Z$ with $X = (X_i)_{i \leq r}$ and $Z = (Z_i)_{i \leq r}$ independent, i.i.d. $\mathcal{N}(0, 1)$. Then

$$S_{xy} - \rho = \rho(S_{xx} - 1) + \underbrace{\sqrt{1 - \rho^2} \frac{1}{r} \sum_{i=1}^r X_i Z_i}_{=: T_r}. \quad (45)$$

From [Vershynin \(2018\)](#) Theorem 6.2.2 page 183, the bilinear Hanson–Wright inequality (for independent sub-Gaussian X, Z and fixed $M \in \mathbb{R}^{r \times r}$) states

$$\mathbb{P}\left(|X^\top M Z| \geq u\right) \leq 2 \exp\left[-c \min\left(\frac{u^2}{K^4 \|M\|_F^2}, \frac{u}{K^2 \|M\|}\right)\right], \quad (46)$$

where $K = O(1)$ is the Orlicz sub-gaussian norm of the coordinates. For $M = I_r$ we have $\|M\|_F^2 = r$ and $\|M\| = 1$, hence with $u = r t$ and $t \in (0, 1)$,

$$\mathbb{P}\left(\left|\frac{1}{r} X^\top Z\right| \geq t\right) \leq 2 \exp(-c r t^2). \quad (47)$$

Thus T_r is sub-Gaussian at rate $\exp(-c r t^2)$:

$$\mathbb{P}(|T_r| \geq t) \leq 2 \exp(-c r t^2). \quad (48)$$

For $S_{xx} = (1/r)\|X\|^2$ and $S_{yy} = (1/r)\|Y\|^2$, Laurent–Massart Gaussian quadratic chaos inequality in (4.1) page 1325 [Laurent and Massart \(2000\)](#) yields, for any $s \in (0, 1)$,

$$\mathbb{P}(|S_{xx} - 1| \geq s) \leq 2 \exp(-c r s^2), \quad \mathbb{P}(|S_{yy} - 1| \geq s) \leq 2 \exp(-c r s^2). \quad (49)$$

Let $f(a, b, c) = a/\sqrt{bc}$. A first-order expansion at $(\rho, 1, 1)$ yields

$$|\hat{\rho}_r - \rho| = |f(S_{xy}, S_{xx}, S_{yy}) - f(\rho, 1, 1)| \leq |S_{xy} - \rho| + \frac{|\rho|}{2} (|S_{xx} - 1| + |S_{yy} - 1|) + R, \quad (50)$$

where the remainder $R = O((|S_{xx} - 1| + |S_{yy} - 1|)^2)$ is negligible on the event $\{|S_{xx} - 1| \vee |S_{yy} - 1| \leq c_0\}$ (for some absolute c_0 , e.g., $1/4$). Choosing $s = \kappa t$ for a sufficiently small absolute $\kappa > 0$, and splitting $|S_{xy} - \rho|$ as in step 1, a union bound gives

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq \underbrace{\mathbb{P}(|T_r| \geq c_1 t)}_{\leq 2e^{-c r t^2}} + \underbrace{\mathbb{P}(|S_{xx} - 1| \geq c_2 t)}_{\leq 2e^{-c r t^2}} + \underbrace{\mathbb{P}(|S_{yy} - 1| \geq c_2 t)}_{\leq 2e^{-c r t^2}} + \mathbb{P}(R \not\leq t). \quad (51)$$

The last term is absorbed by adjusting κ (small-probability event controlled by the same inequalities). Therefore, for absolute constants $c, C > 0$,

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq C \exp(-c r t^2), \quad t \in (0, 1). \quad (52)$$

■

Remark 39 (“Bessel” variant: sum of Gaussian products) *Each product $X_i Z_i$ has a symmetric sub-exponential tail (density involving the Bessel function K_0). By Bernstein’s inequality for i.i.d. sub-exponential variables, $\frac{1}{r} \sum_{i=1}^r X_i Z_i$ satisfies the same $\exp(-c r t^2)$ bound for $t \in (0, 1)$, yielding an alternative proof of the bilinear step.*

By Theorem 22, for $\epsilon/(2L_w) \in (0, 1)$:

$$\mathbb{P}\left(|\hat{w}_r - 1| \geq \frac{\epsilon}{2L_w}\right) \leq 4 \exp\left(-\frac{r \epsilon^2}{32L_w^2}\right). \quad (53)$$

Combining both terms:

$$\begin{aligned} \mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) &\leq \underbrace{\mathbb{P}\left(|\hat{\rho}_r - \rho| \geq \frac{\epsilon}{2L_u}\right)}_{\leq C \exp\left(-\frac{c r \epsilon^2}{4L_u^2}\right)} + \underbrace{\mathbb{P}\left(|\hat{w}_r - 1| \geq \frac{\epsilon}{2L_w}\right)}_{\leq 4 \exp\left(-\frac{r \epsilon^2}{32L_w^2}\right)} \end{aligned} \quad (54)$$

so that

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq C \exp\left(-\frac{c r \epsilon^2}{4L_u^2}\right) + 4 \exp\left(-\frac{r \epsilon^2}{32L_w^2}\right). \quad (55)$$

Taking $C_1 = C + 4$ and $C_2 = \min(c/(4L_u^2), 1/(32L_w^2))$, we obtain

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq C_1 \exp(-C_2 r \epsilon^2). \quad (56)$$

■

B.4. Proof of Corollary 27: Mean NTK over Fisher–Kibble has same RKHS

Proof We prove that under Assumption 2.1, the mean NTK $\tilde{\Theta}^{(2)}$ for the three-layer case (obtained by taking expectations over Fisher and Kibble distributions) is a zonal kernel that induces the same RKHS as the shallow ReLU kernel.

Puiseux expansion of mean NTK near $\rho = 1$: The key observation is that the mean NTK $\tilde{\Theta}^{(2)}(\rho)$ is a zonal kernel (depends only on ρ). Near $\rho = 1$ (for $t \geq 0$), we need to compute the Puiseux expansion of $\mathbb{E}[\arccos(\hat{\rho}_r)]$ when $\rho = 1 - t$, where $\hat{\rho}_r \sim \text{Fisher}(1 - t, r)$.

This is highly non-trivial because we must compute:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_{-1}^1 \arccos(u) p_{\text{Fisher}}(u \mid \rho = 1 - t, r) du, \quad (57)$$

where the full Fisher distribution density is:

$$p_{\text{Fisher}}(u \mid \rho, r) = \frac{(r-2) \Gamma(r-1) (1-\rho^2)^{\frac{r-1}{2}} (1-u^2)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1-\rho u)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho u}{2}\right), \quad (58)$$

for $r > 2$, with ${}_2F_1$ the hypergeometric function.

Making the change of variables $v = 1 - u$ and $s = 1 - t$, so $\rho = s = 1 - t$ and $u = 1 - v$, we have:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^2 \arccos(1-v) p_{\text{Fisher}}(1-v \mid s, r) dv. \quad (59)$$

Near $v = 0$ and $t = 0$, we analyze the asymptotic behavior of the Fisher density. For $\rho = s = 1 - t$ and $u = 1 - v$ with $t, v \rightarrow 0^+$:

$$1 - \rho^2 = 1 - (1-t)^2 = 2t - t^2 = 2t(1-t/2), \quad (60)$$

$$1 - u^2 = 1 - (1-v)^2 = 2v - v^2 = 2v(1-v/2), \quad (61)$$

$$1 - \rho u = 1 - (1-t)(1-v) = t + v - tv = t + v + O(tv). \quad (62)$$

The Fisher density near $v = 0$ and $t = 0$ behaves as:

$$p_{\text{Fisher}}(1-v \mid 1-t, r) \sim \frac{(r-2) \Gamma(r-1) (2t)^{\frac{r-1}{2}} (2v)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (t+v)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; 1-\frac{t+v}{2}\right). \quad (63)$$

For the hypergeometric function near its argument $1 - (t+v)/2 \rightarrow 1^-$, we invoke the connection formula in DLMF §15.8.2 [Olver et al. \(2024\)](#) together with the fact that $c - a - b = r - \frac{3}{2} > 0$ for $r \geq 2$ and is a half-integer (hence no logarithmic term appears). Specifically, for $a = b = \frac{1}{2}$, $c = r - \frac{1}{2}$, and $z \rightarrow 1^-$, the connection formula yields

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c) \Gamma(c-a-b)}{\Gamma(c-a) \Gamma(c-b)} + O((1-z)^{c-a-b}). \quad (64)$$

Setting $z = 1 - \frac{t+v}{2}$ we obtain, uniformly for $t, v \rightarrow 0^+$,

$${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; 1-\frac{t+v}{2}\right) = C_1(r) + O((t+v)^{r-\frac{3}{2}}), \quad (65)$$

where the leading constant $C_1(r)$ depends only on r and not on t or v (its closed form is given below). Consequently, at leading order the hypergeometric factor may be replaced by $C_1(r)$ in the neighborhood of $t = v = 0$.

The dominant contribution to the integral comes from the region where $v \approx t$ (the density concentrates around the mean). Near this region, $\arccos(1-v) = \sqrt{2v} + O(v^{3/2})$, so:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^\infty \sqrt{2v} \cdot \frac{(r-2) \Gamma(r-1) (2t)^{\frac{r-1}{2}} (2v)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (t+v)^{r-\frac{3}{2}}} C_1(r) dv + O(t^{3/2}). \quad (66)$$

Making the substitution $w = v/t$, we have $v = tw$, $dv = t dw$, and $t + v = t(1+w)$:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^\infty \sqrt{2tw} \cdot \frac{(r-2) \Gamma(r-1) (2t)^{\frac{r-1}{2}} (2tw)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (t(1+w))^{r-\frac{3}{2}}} C_1(r) \cdot t dw + O(t^{3/2}). \quad (67)$$

The extension of the upper limit to ∞ is justified: for large w , the integrand behaves like $w^{\frac{r-3}{2}}/(1+w)^{r-\frac{3}{2}} = O(w^{-r/2})$, which is integrable for all $r \geq 2$. Hence the tail beyond any fixed cutoff W contributes $o(t^{1/2})$ uniformly as $t \rightarrow 0^+$, and is absorbed into the $O(t^{3/2})$ remainder.

Simplifying the powers of t :

$$= \sqrt{2t} \cdot t^{\frac{r-1}{2} + \frac{r-4}{2} + 1 - (r-\frac{3}{2})} \cdot \frac{(r-2) \Gamma(r-1) (2)^{\frac{r-1}{2}} (2w)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1+w)^{r-\frac{3}{2}}} C_1(r) \cdot \sqrt{w} dw + O(t^{3/2}), \quad (68)$$

where the exponent is $\frac{r-1}{2} + \frac{r-4}{2} + 1 - (r-\frac{3}{2}) = 0$, so:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} \cdot I(r) + O(t^{3/2}), \quad (69)$$

where

$$I(r) = \int_0^\infty \sqrt{w} \cdot \frac{(r-2) \Gamma(r-1) (2)^{\frac{r-1}{2}} (2w)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1+w)^{r-\frac{3}{2}}} C_1(r) dw. \quad (70)$$

Evaluating the integral yields the exact constant

$$I(r) = \frac{(r-2) 2^{r-\frac{5}{2}} \Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2}-1)}{\sqrt{2\pi} \Gamma(r-1)} C_1(r), \quad C_1(r) = \frac{\Gamma(r-\frac{1}{2}) \Gamma(r-\frac{3}{2})}{\Gamma(r-1)^2}. \quad (71)$$

Therefore,

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} I(r) + O(t^{3/2}). \quad (72)$$

Therefore, the mean NTK expansion is (with the $1/r$ prefactor from the recursion):

$$\tilde{\Theta}^{(2)}(1-t) = 1 + \frac{1}{r} (K_\infty(1) - 1) - \frac{1}{r} \left[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r) \right] t^{1/2} + O(t^{3/2}). \quad (73)$$

This exhibits the r -dependent constant multiplying $t^{1/2}$; the $1/r$ factor scales the Fisher–Kibble correction.

Remark 40 (Closed form via Beta function and asymptotics) *Using the change of variables $w = v/t$ and Euler’s Beta integral, the inner integral appearing in $I(r)$ can be evaluated explicitly:*

$$\int_0^\infty \frac{w^{\frac{r-3}{2}}}{(1+w)^{r-\frac{3}{2}}} dw = B\left(\frac{r-1}{2}, \frac{r}{2}-1\right) = \frac{\Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2}-1)}{\Gamma(r-\frac{3}{2})}. \quad (74)$$

Substituting this identity into the pre-factors gives the stated closed form for $I(r)$ in (75) below, which matches the expression above:

$$I(r) = \frac{(r-2) 2^{r-\frac{5}{2}}}{\sqrt{2\pi}} \frac{\Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2}-1)}{\Gamma(r-1)} C_1(r), \quad C_1(r) = \frac{\Gamma(r-\frac{1}{2}) \Gamma(r-\frac{3}{2})}{\Gamma(r-1)^2}. \quad (75)$$

The asymptotic scaling $I(r) \sim C/\sqrt{r}$ as $r \rightarrow \infty$ is established in Proposition 41 (Appendix B.5). In particular, the leading Puiseux coefficient

$$1 - \left[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r) \right] t^{1/2} + O(t^{3/2}) \quad (76)$$

is strictly positive and depends smoothly on r . This confirms that the endpoint exponent $\frac{1}{2}$ (hence the RKHS spectral decay rate) is unchanged by the Fisher–Kibble expectation, while the multiplicative constant is an $O(1)$ perturbation of the shallow ReLU constant $2/\pi$.

RKHS equivalence via endpoint behavior. The mean NTK $\tilde{\Theta}^{(2)}(\rho)$ is a zonal kernel with a Puiseux expansion that differs from the deterministic kernel $K_\infty(\rho)$ due to the Fisher–Kibble expectations. However, the leading-order behavior near $\rho = 1$ is preserved: both kernels have the same $t^{1/2}$ leading term in their Puiseux expansions.

By Theorem 1 of Bietti and Bach (2021), the RKHS spectral decay is determined by the leading-order Puiseux expansion coefficient (the $t^{1/2}$ term). Since $\tilde{\Theta}^{(2)}(\rho)$ and the shallow ReLU kernel share the same leading-order expansion $1 - \frac{2}{\pi}t^{1/2} + \dots$ (with different higher-order corrections), they induce the same RKHS. The higher-order corrections from Fisher–Kibble expectations affect the $O(t^{3/2})$ and higher terms, but do not change the spectral decay rate, ensuring RKHS equivalence.

Extension to general L : open problem. The extension to general depth $L \geq 3$ remains open. A rigorous proof would require: (i) the conditional law of the Fisher chain $(\rho, \rho_1, \dots, \rho_{L-2})$ given $\rho_{L-1} = u$, or equivalently the $(L-1)$ -fold marginal distribution of ρ_{L-1} given ρ ; (ii) an explicit Laplace-type expansion of the integral $\int_{-1}^1 \dot{\Sigma}^{(L)}(u) \mathbb{E}[\Theta^{(L-1)} \mid \rho_{L-1} = u] p_{\text{Fisher}}(u \mid \rho, r) du$ (and analogous terms) as $\rho = 1 - t \rightarrow 1^-$. The conditional mean $\mathbb{E}[\Theta^{(L-1)} \mid \rho_{L-1} = u]$ is not equal to $\tilde{\Theta}^{(L-1)}(u)$, since the former conditions on the chain endpoint while the latter fixes the input; verifying the Puiseux structure of the conditional expectation would require a detailed analysis of the backward Fisher-chain density. We conjecture that the same $t^{1/2}$ leading exponent holds for all L , and that the mean NTK induces the same RKHS; we leave a complete proof for future work. ■

B.5. Asymptotic scaling of $I(r)$

The Fisher–Kibble integral $I(r)$ defined in (75) governs the r -dependent coefficient in the mean NTK’s Puiseux expansion near $\rho = 1$. The following proposition gives its scaling as $r \rightarrow \infty$.

Proposition 41 ($I(r)$ decays as $1/\sqrt{r}$) Let $I(r)$ be as in (75) with $C_1(r) = \Gamma(r - \frac{1}{2})\Gamma(r - \frac{3}{2})/\Gamma(r - 1)^2$. Then

$$I(r) \sim \frac{C}{\sqrt{r}} \quad \text{as } r \rightarrow \infty, \quad \text{for some } C > 0, \quad (77)$$

i.e., $I(r) = O(1/\sqrt{r})$. For fixed $r \geq 3$, $I(r)$ is bounded and strictly positive.

Proof Using the Beta representation $B((r-1)/2, r/2-1) = \Gamma((r-1)/2)\Gamma(r/2-1)/\Gamma(r-3/2)$, the closed form reduces to $I(r) = (r-2) 2^{r-5/2}/\sqrt{2\pi} \cdot B \cdot \Gamma(r-3/2)\Gamma(r-1/2)/\Gamma(r-1)^3$. By Stirling’s formula and the asymptotics $B(a, b) \sim \sqrt{2\pi} a^{a-1/2} b^{b-1/2}/(a+b)^{a+b-1/2}$ when $a, b \rightarrow$

∞ with $a \sim b \sim r/2$, one has $B((r-1)/2, r/2-1) \sim \sqrt{2\pi} 2^{-r}/\sqrt{r}$. The Gamma-ratio $\Gamma(r-1/2)\Gamma(r-3/2)/\Gamma(r-1)^2 \sim 1$. Combining these yields $I(r) \sim C/\sqrt{r}$. $\blacksquare$

Implications: (i) the Puiseux coefficient $c_1(r) = -[2/\pi + (\sqrt{2}/(2\pi)) I(r)]$ satisfies $c_1(r) \rightarrow -2/\pi$ as $r \rightarrow \infty$, recovering the shallow ReLU constant; (ii) the Fisher–Kibble correction to the mean NTK is $O(1/(r\sqrt{r}))$ in the $t^{1/2}$ term when $I(r) \sim C/\sqrt{r}$.

Appendix C. Spectrum of 3-layers Gram NTK Matrix

C.1. Asymptotic Signal-Noise Decomposition

Proof [Proof of Theorem 32] We establish the asymptotic distribution of the empirical kernel $\hat{\Theta}_r^{(2)} = \Psi(\hat{\rho}_r, \hat{w}_r)$ via a joint CLT and the multivariate Delta method. Let $\mathbf{Z}_r = (\hat{\rho}_r, \hat{w}_r)^\top$. For isotropic Gaussian projections in rank r , Fisher’s theory [Fisher \(1915\)](#); [Hotelling \(1953\)](#) gives

$$\sqrt{r}(\hat{\rho}_r - \rho) \xrightarrow{d} \mathcal{N}(0, (1 - \rho^2)^2). \quad (78)$$

Writing $u = \|x\|^2$, $v = \|y\|^2$ and $\hat{w}_r = \sqrt{uv}/r$, Kibble’s bivariate chi-square yields

$$\sqrt{r}(\hat{w}_r - 1) \xrightarrow{d} \mathcal{N}(0, 1 + \rho^2). \quad (79)$$

Angular and radial components are independent for isotropic Gaussians, hence

$$\sqrt{r}(\mathbf{Z}_r - (\rho, 1)^\top) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{\Lambda}(\rho)), \quad \mathbf{\Lambda}(\rho) = \begin{pmatrix} (1 - \rho^2)^2 & 0 \\ 0 & 1 + \rho^2 \end{pmatrix}. \quad (80)$$

The map $\Psi(u, w) = \Theta^{(1)}(u)(1 - \arccos(u)/\pi) + w \Sigma^{(1)}(u) + 1$ is C^1 near $(\rho, 1)$. Therefore,

$$\sqrt{r}(\Psi(\mathbf{Z}_r) - \Psi(\rho, 1)) \xrightarrow{d} \mathcal{N}(0, \nabla \Psi(\rho, 1)^\top \mathbf{\Lambda}(\rho) \nabla \Psi(\rho, 1)). \quad (81)$$

We have $\partial_u \Psi(\rho, 1) = \partial_\rho K_\infty(\rho)$ and $\partial_w \Psi(\rho, 1) = \Sigma^{(1)}(\rho)$. Using the diagonal covariance, we obtain

$$\sigma_{\text{tot}}^2(\rho) = (\partial_\rho K_\infty(\rho))^2 (1 - \rho^2)^2 + (\Sigma^{(1)}(\rho))^2 (1 + \rho^2). \quad (82)$$

Hence

$$\hat{\Theta}_r^{(2)} = K_\infty(\rho) + \frac{1}{\sqrt{r}} \mathcal{G}(\rho) + o_p(r^{-1/2}), \quad \mathcal{G}(\rho) \sim \mathcal{N}(0, \sigma_{\text{tot}}^2(\rho)), \quad (83)$$

which proves the claim. $\blacksquare$

C.2. Proof of Theorem 33: Deformed Marchenko–Pastur with Spike

Proof We prove Theorem 33 by a direct application of [El Karoui \(2010, Theorem 2.1\)](#), which describes the limiting spectrum of high-dimensional *inner-product kernel matrices*.

Setting (deterministic kernel matrix). Let $x_1, \dots, x_n \in \mathbb{R}^d$ be generated as in Assumption 3 with $d \rightarrow \infty$ and $n/d \rightarrow \gamma_{\text{ratio}} \in (0, \infty)$. Let $X = [x_1, \dots, x_n] \in \mathbb{R}^{d \times n}$. Define the (uncentered) kernel Gram matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$ by

$$\mathbf{M}_{ij} = f\left(\frac{x_i^\top x_j}{d}\right), \quad f(t) = K_\infty(t),$$

where $K_\infty(\cdot)$ is the deterministic three-layer kernel function defined in the main text. By the smoothness discussion in Section 4 and the regularity conditions in Assumption 3, f satisfies the differentiability hypotheses required by El Karoui (2010).

El Karoui linearization. El Karoui (2010, Theorem 2.1) implies that $\mathbf{M}$ is asymptotically equivalent in operator norm to a spike-plus-Wishart-plus-shift matrix of the form

$$\mathbf{K} = \alpha \mathbf{1}\mathbf{1}^\top + \beta \frac{X^\top X}{d} + \gamma I_n,$$

where the coefficients are determined by f and the trace limit $\ell = \lim_{d \rightarrow \infty} \text{trace}(\Sigma_d)/d$:

$$\alpha = f(0) = K_\infty(0), \quad \beta = f'(0) = K'_\infty(0), \quad \gamma = f(\ell) - f(0) - \ell f'(0) = K_\infty(\ell) - K_\infty(0) - \ell K'_\infty(0).$$

The rank-one term $\alpha \mathbf{1}\mathbf{1}^\top$ yields a single eigenvalue of order n (the spike). The remaining $n - 1$ eigenvalues are those of the shifted sample covariance term $\beta X^\top X/d + \gamma I_n$, whose empirical distribution converges to a (deformed) Marchenko–Pastur law with support $[\gamma + \beta(1 - \sqrt{\gamma_{\text{ratio}}})^2, \gamma + \beta(1 + \sqrt{\gamma_{\text{ratio}}})^2]$ under Assumption 3. This matches the statement of Theorem 33.

Scope note. The theorem concerns the deterministic kernel matrix $f(x_i^\top x_j/d)$. Incorporating additional entrywise randomness (e.g. finite- r Fisher–Kibble fluctuations or finite-width effects) into the limiting ESD requires further resolvent/local-law inputs and is outside the scope of this appendix. ■